Boosting the Generalization Ability for Hyperspectral Image Classification Using Spectral–Spatial Axial Aggregation Transformer

Enzhe Zhao, *Student Member, IEEE*, Zhichang Guo, Shengzhu Shi, Yao Li, *Member, IEEE*, Jia Li, and Dazhi Zhang

***Abstract*— In the hyperspectral image classification (HSIC) task, the most commonly used model validation paradigm is partitioning the training–test dataset through pixelwise random sampling. By training on a small amount of data, the deep learning model can achieve almost perfect accuracy. However, in our experiments, we found that the high accuracy was reached because the training and test datasets share a lot of information. On nonoverlapping dataset partitions, well-performing models suffer significant performance degradation. To this end, we propose a spectral–spatial axial aggregation transformer model, namely, SaaFormer, which preserves generalization across dataset partitions. SaaFormer applies a multilevel spectral extraction structure to segment the spectrum into multiple spectrum clips such that the wavelength continuity of the spectrum across the channel is preserved. For each spectrum clip, the axial aggregation attention mechanism, which integrates spatial features along multiple spectral axes, is applied to mine the spectral characteristic. The multilevel spectral extraction and the axial aggregation attention emphasize spectral characteristics to improve the model generalization. The experimental results on five publicly available datasets demonstrate that our model exhibits comparable performance on the random partition while significantly outperforming other methods on nonoverlapping partitions. Moreover, SaaFormer shows excellent performance on background classification.**

***Index Terms*— Axial aggregation, generalization, hyperspectral image classification (HSIC), spectral–spatial, transformer.**

I. Introduction

HYPERSPECTRAL remote sensing (RS) images are a type of satellite or airborne images that capture data across a wide range of wavelengths, which is concerned with the extraction of meaningful information based on the radiance acquired by the sensor at short or long distances [1]. Unlike multispectral images that capture data in a limited number of wavelength bands, HS images record information across hundreds of narrow bands. In contrast to other RS images, HS image enables a detailed and precise analysis of Earth's surface due to variations in how different materials and surfaces reflect light across the electromagnetic spectrum [2], [4]. Consequently, a profound understanding and thorough modeling are critical for HS image analysis.

In the early stages of HS image analysis, researchers modeled the spectra for individual pixels, and traditional classifiers were commonly employed. They assumed that spectra from the same type of area followed the same distribution [5], [6], were geometrically close to each other [7], or shared the same latent feature space [8]. Typically, these techniques involve a two-stage process. Initially, they reduce the dimensionality of the HSI data and extract informative features. Subsequently, spectral classifiers utilize these features for classification purposes [3], such as the k-nearest neighbor [9], the Bayesian estimation method [10], the multinomial logistic regression [11], and the support vector machine (SVM) [12]. Furthermore, various approaches have been devised to accomplish dimension reduction and extract spectral information. These include techniques such as principal component analysis (PCA) [13], independent component analysis (ICA) [14], and linear discriminant analysis (LDA) [15]. However, these methods ignore the spatial correlation among the pixels in the spatial dimension, and due to factors such as changes in illumination or surface conditions, the spectra can vary, diminishing the generalizability of modeling spectra from a single pixel to HS images. To enhance the extraction of spatial features from images, researchers have introduced several mathematical morphological approaches, including extended MP (EMP) [16] and extended multiattribute profile (EMAP) [17].

The rapid advancement of deep learning technology has hastened the progress of image processing techniques across various domains and has also contributed to the technical innovation of RS image processing [18]. For HSI, a multitude of diverse classification methods employing deep models has been put forward, such as 1-D convolutional neural networks (CNNs) (1DCNNs) [19], 2DCNNs [20], [21], [22], [23], 3DCNNs [24], [25], and recurrent neural network (RNN) [26]. Hu et al. [19] introduced a 1DCNN classifier to identify

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The authors are with the Department of Computational Mathematics, School of Mathematics, Harbin Institute of Technology, Harbin 150001, China (e-mail: mathssz@hit.edu.cn).

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and incorporate local dependencies between adjacent spectral bands enhancing the overall comprehension and interpretation of HS images. He et al. [25] presented the M3D-CNN model for HS image classification, allowing for joint modeling of multiple spectral bands and enabling the integration and fusion of diverse features across different scales. Roy et al. [27] introduced a hierarchical network structure that combines 3-D and 2-D CNN characteristics to efficiently extract spatial–spectral features, reducing computational complexity and improving classification accuracy.

With the prevalence of residual networks in the field of image classification, Zhong et al. [28] introduced a spatial–spectral residual network for HSI classification, leveraging features from multiple layers to enhance feature utilization. Meanwhile, in [29], a pyramid residual network was devised, enhancing feature map dimensionality and extracting information efficiently through grouped residual blocks. Roy et al. [30] investigated an improved spectral–spatial residual network for joint feature extraction. Besides, they utilized a lightweight paradigm, employing the squeeze-and-excitation ResNet for spatial and spectral feature extraction and integrating a bag-of-features learning mechanism to achieve accurate final classification results [31]. Meanwhile, in order to fully leverage CNNs, various complex structures have been introduced to achieve efficient classification, such as Rotation equivariant CNNs [32] and gradient centralized convolutions [33], [34]. Although CNNs are effective at capturing spatial contextual information, they struggle to integrate long- and middle-term dependencies sequentially. They may be more suitable for handling local spatial features rather than spectral features, thus failing to fully capture the subtle differences and key features in the spectral information. Consequently, their performance in classifying HSI might suffer, especially when dealing with classes sharing similar spectral signatures, which complicates the extraction of distinguishing spectral attributes.

On the other hand, recursive neural networks (RNNs) can be employed to model the spectral signatures in HSIs, which can construct a sequence model to effectively simulate the relationship between adjacent spectral bands [26]. Yet, for longer hyperspectral sequences, RNNs may encounter the issue of vanishing or exploding gradients, resulting in training difficulties and reduced computational efficiency. Thus, we consider the aforementioned limitations by rethinking HSI classification using transformers [35], in which, as leading-edge backbone architectures, transformers utilize self-attention mechanisms to optimize the processing and analysis of sequential data. Its integration seamlessly fuses spatial and spectral information through self-attention, enriching the classification model's understanding of holistic features. Hong et al. [36] developed a new model, SpectralFormer, which can learn spectral representation information from groupwise neighboring bands and construct a cross-layer transformer encoder module. He et al. [37] proposed a bidirectional encoder representation for a transformer that incorporates flexible and dynamic input regions for pixel-based classification of HSIs. Roy et al. [38] introduced morphFormer, leveraging learnable spectral and spatial morphological convolution operations combined with attention mechanisms for improved performance. Sun et al. [39] implemented spatial and spectral tokenization in the encoder to extract local spatial information and establish long-range relations between neighboring sequences, while transformer-based models are computationally expensive when dealing with long sequential data due to the self-attention across the entire sequence. This complexity leads to high computational costs and memory consumption. In addition, the sparsity and redundancy of HSIs pose a challenge to the self-attention mechanism to capture crucial information. Consequently, these models may struggle to extract features and preserve spectral information.

Furthermore, recent advances in RS technology have increased the availability of multisensor data, allowing for multiple representations of the same geographical region. Multimodal data integration has also attracted significant attention [40], [41], [42], [43]. Hong et al. [44] Wu et al. [45] proposed a general and unified multimodal deep learning framework focusing on RS image classification. This framework integrates pixel-level labeling guided by an FC design and spatial–spectral joint classification with a CNN framework. Roy et al. [41] developed a multimodal fusion transformer to extract features from HSIs and fuse them with a CLS token derived from light detection and ranging (LiDAR) data to enhance the joint classification performance. They further [46], [47] contributed to the benchmark dataset, transfer learning, and plug-and-play techniques for the multimodal RS. Recently, with the significant surge in foundation models based on pretraining techniques [48], SpectralGPT [49] pioneered a universal RS base model utilizing a 3-D generative pretrained transformer (GPT).

Meanwhile, with the development of artificial intelligence (AI) and the emergence of large-scale remote sensing image datasets, the trustworthy AI in RS has become a focal point of widespread attention [50]. Currently, the development of RS image data presents multimodal features, such as multiangle, multitemporal, multiplatform, and multiscale characteristics [40], [51]. Multimodal data-driven RS intelligent interpretation algorithms (RSIIAs) have been widely used in various fields of RS image processing. At present, data-driven AI has advanced notably in RS, yet it has introduced challenges regarding interpretability and trustworthy. To address this, researchers have focused on investigating trust at two levels—data and model—within the RS field. On the one hand, to enhance the trustworthiness of RS at the data level, key strategies involve improving data quality, integrating multimodal data, updating data, and promoting data sharing. Spatial–temporal–spectral data fusion [52] facilitates obtaining RS imagery with high temporal, spatial, and spectral resolutions. The fusion of multisource and multimodal data [53] provides additional information gain for RS intelligent interpretation. In terms of the AI models, given their black-box characteristics, research on model interpretability is crucial in assessing the credibility of the models. Kakogeorgiou and Karantzalos [54] explored interpretable AI methods applied to the task of multilabel classification of RS. Moreover, assessing model uncertainty is a crucial approach to improving the reliability of AI. He et al. [55] addressed data and model uncertainty

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in HSI classification, leading to the development of a reliable HSI classifier through the evaluation and mitigation of these uncertainties. Wang et al. [56] introduced a technique for quantifying the uncertainty of multimodal data in order to enhance the precision and trustworthiness of data fusion. In summary, trustworthy research in the data-driven RS is extensive, emerging a series of AI methods for measuring, evaluating, and enhancing the trustworthiness of RS interpretation.

Based on the above analysis and discussion, a new spectral–spatial fusion transformer model is introduced in this work to take advantage of the transformer's ability to obtain local spatial semantic information and model the relationship between adjacent sequences. The objective of our model, SaaFormer, is to extract and fuse spectral–spatial features across multiple scales along the spectral dimension from the patch embeddings of HSI, as well as to ensure efficient computation and grasp intricate spectral nuances, generating comprehensive feature representations, and bolstering the generalization and credibility of the model.

The primary contributions of this article can be outlined as follows.

1) We provide a spectral–spatial axial aggregation transformer framework, namely, SaaFormer, which emphasizes fine-grained feature extraction through multispectral extraction modules and efficiently aggregates global spatial and spectral feature information with axial transformer modules, thereby reducing computational complexity while significantly improving the handling of redundancy in HSIs and enhancing the capability to extract and maintain spectral features. Consequently, it has achieved state-of-the-art performance across five public datasets.
2) We have raised significant concerns regarding the generalization of HSI classification. Through comprehensive theoretical and experimental analyses of information sharing, we discovered that excessive overlap between the training set and test set tends to yield overly optimistic model performance. Our model emphasizes central pixels with axial transformer modules, thereby mitigating interference from related pixels more effectively and improving generalization capabilities, as validated by various partitioning methods.
3) The proposed SaaFormer achieves exceptional performance in the background classification task, which further supports its superior generalization capabilities and performs well across diverse datasets and varying conditions.

The remaining structure of the article is outlined as follows. Section II presents the model structure that we have proposed along with its key details. In Section III, we discuss several dataset partitioning methods aimed at mitigating data share information issues. In Section IV, we conduct extensive experiments, offering specific experimental data and visual results. Finally, Section V offers comprehensive conclusions based on a thorough analysis.

II. Methodology

A. Architecture of SaaFormer

The architecture of the proposed SaaFormer is shown in Fig. 1. SaaFormer consists of two primary components: the multilevel spectral extraction structure and the axial aggregation attention mechanism. The multilevel spectral extraction structure conducts K different levels of segmentation plans to the linear embedding $\mathbf{x} \in \mathbb{R}^{h\times w\times c}$ of the sample $\mathbf{x}_0 \in \mathbb{R}^{h\times w\times c}$ such that $\mathbf{x}$ is segmented into clips with length $c_1, c_2, \ldots, c_K$. Segmenting the spectra into clips instead of weighting each spectra channel can preserve the physical continuity of the spectra wavelength across channels. Extracting features in the unit of spectra clips regroups the information in each single spectra channel and improves model generalization ability. The multilevel segmentation plans ensure that the multilevel features can be observed from clips in different granularities.

For each spectrum clip, the axial aggregation transformer block is applied to mine the spectral characteristic. The axial transformer block integrates row and columnwise spatial features along multiple spectral axes to emphasize spectral characteristics. Then, the features mined from each spectrum clip are integrated as the nonlinear embedding of the sample. The axial attention mechanism applies the spatial information as the auxiliary of the spectra information analysis task such that the model can have a deep understanding of the spectra instead of simply memorizing spatial features to improve the model generalization. The implementation details of the multilevel spectral extraction structure and the axial aggregation attention mechanism are described in Sections II-B and II-C.

B. Multilevel Spectral Extraction

From the view of the absorption or reflection characteristics of different materials for different wavelengths of light, the absorption peaks and the reflection peaks are usually interspersed occasionally. Thus, we propose to analysis the spectra of HS images with multilevel spectra clips. In the multilevel spectral–spatial extraction structure, the linear embeddings of the samples are segmented into nonoverlapping continuous spectral segments of varying lengths, which are considered spectral clips. As illustrated in Fig. 1, the feature map x is divided into spectral clips of different lengths (i.e., $c_1, c_2, \ldots, c_K$) in spectral dimensions. The multilevel segmentation plans ensure that the multilevel features are observed from clips at different levels of granularity. The spectral characteristic of each spectrum clip is mined using the axial aggregation transformer block.

C. Axial Transformer Block

The axial transformer block is applied to each spectra clip for a deep understanding of the spectra segmentation. It utilizes spatial information as an auxiliary by projecting the spatial information onto spectral axes. To reduce the structure redundancy, we project the rowwise spatial information and the columnwise spatial information separately to the spectral axes. Then, the spatial-enhanced spectral features are integrated through the multihead self-attention mechanism

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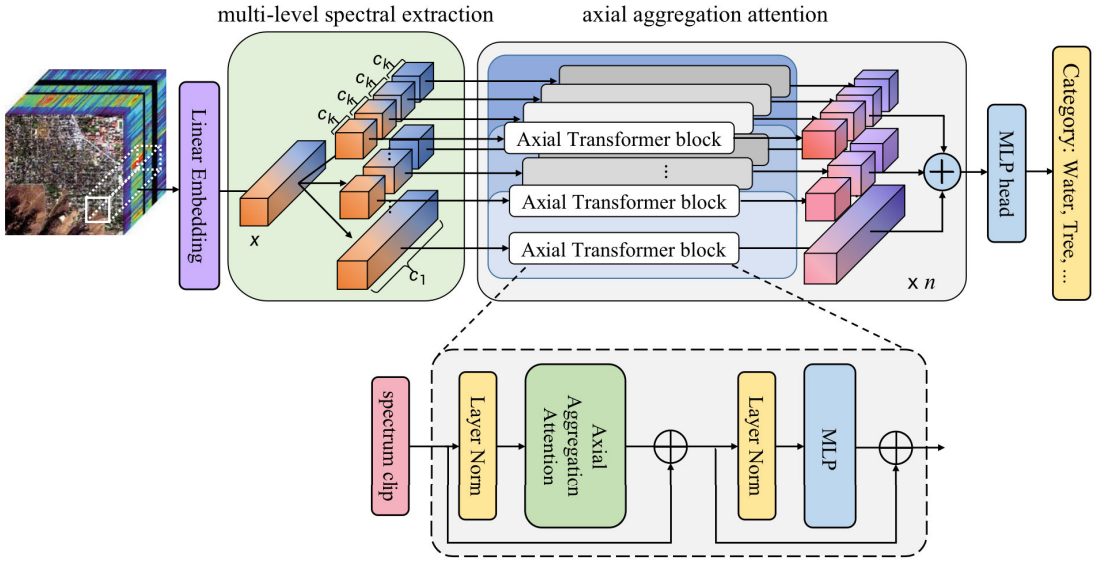


Fig. 1. Overview architecture of SaaFormer for the HS image classification task. The model architecture consists of two primary components: the multilevel spectral extraction structure and the axial aggregation attention.

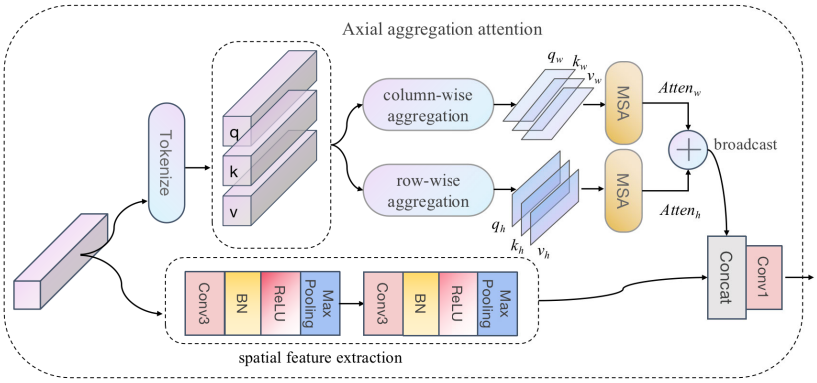


Fig. 2. Diagram of the axial transformer block in the proposed SaaFormer. The attention process compresses the feature map either vertically or horizontally and calculates self-attention separately for each axis.

(MSA), which provides adaptive attention to different spectral channels. The axial self-attention mechanism considers the interrelationships among spectral bands at each pixel location and aggregating features in the spatial domain to improve the model generalization capability. The axial attention mechanism has also been used in various tasks [57], [58], [59], [60] as lightweight modifications to improve the self-attention mechanism [61], [62], [63]. Hence, in addressing the challenge of HSIC, we improve [60]. Due to its imperfect alignment with the specific characteristics of high-dimensional spectral data, we have undertaken further improvements by designing the spatial–spectral axial aggregation attention mechanism.

The structure details are shown in Fig. 2. First, the query, key, and value of the spectral clip $\mathbf{x} \in \mathbb{R}^{h\times w\times c}$ are calculated through linear projections, i.e., $\mathbf{q} = \mathbf{x}\mathbf{W}_q$, $\mathbf{k} = \mathbf{x}\mathbf{W}_k$, and $\mathbf{v} = \mathbf{x}\mathbf{W}_v$, where $\mathbf{W}_q, \mathbf{W}_k, \mathbf{W}_v \in \mathbb{R}^{c\times d}$ are all learnable matrices and d is the given hyperparameter. Then, we achieve horizontal and vertical spatial information projection by taking the maximum value of the query feature map in the row and column directions, $\mathbf{q}_h = \max_w(\mathbf{q}), \mathbf{q}_w = \max_h(\mathbf{q})$. The aggregation operation consolidates global information by compressing it into a single axis, and the operation on $\mathbf{q}$ also repeats on $\mathbf{k}$ and $\mathbf{v}$ to obtain $\mathbf{q}_h, \mathbf{k}_h, \mathbf{v}_h \in \mathbb{R}^{h\times d}$ and $\mathbf{q}_w, \mathbf{k}_w, \mathbf{v}_w \in \mathbb{R}^{w\times d}$. The last addition in (1) is realized by the broadcast operation, i.e., $\mathbf{z} \in \mathbb{R}^{h\times w\times d}$

$$\begin{aligned} \text{Atten}_h &= \text{SoftMax}\left(\mathbf{q}_h\mathbf{k}_h^T/\sqrt{d}\right)\mathbf{v}_h \\ \text{Atten}_w &= \text{SoftMax}\left(\mathbf{q}_w\mathbf{k}_w^T/\sqrt{d}\right)\mathbf{v}_w \\ \mathbf{z} &= \text{Broadcast}(\text{Atten}_h + \text{Atten}_w). \end{aligned} \tag{1}$$

To introduce the position information [58], [60], [61], [64], or equivalently the wavelength information, to the features, we include a learnable relative position encoded to each head, which is gained from learnable parameters $Pe \in \mathbb{R}^{N\times d}$, and N changes with the input parameters. Apply the learnable relative position encode as $\mathbf{p}_h^q, \mathbf{p}_h^k, \mathbf{p}_h^v \in \mathbb{R}^{h\times d}$ directly adding to $\mathbf{q}_h, \mathbf{k}_h, \mathbf{v}_h$, and $\mathbf{p}_w^q, \mathbf{p}_w^k, \mathbf{p}_w^v \in \mathbb{R}^{w\times d}$ are also applied to $\mathbf{q}_w, \mathbf{k}_w, \mathbf{v}_w \in \mathbb{R}^{w\times d}$ in the same way. Thus, the

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positional-aware aggregation attention can be expressed as

$$\begin{aligned}\text{Atten}_h &= \text{SoftMax}\left(\frac{\left(\mathbf{q}_h + \mathbf{p}_h^q\right)\left(\mathbf{k}_h + \mathbf{p}_h^k\right)^T}{\sqrt{d}}\right)\left(\mathbf{v}_h + \mathbf{p}_h^v\right)\\ \text{Atten}_w &= \text{SoftMax}\left(\frac{\left(\mathbf{q}_w + \mathbf{p}_w^q\right)\left(\mathbf{k}_w + \mathbf{p}_w^k\right)^T}{\sqrt{d}}\right)\left(\mathbf{v}_w + \mathbf{p}_w^v\right)\\ \mathbf{z} &= \text{Broadcast}(\text{Atten}_h + \text{Atten}_w). \end{aligned} \tag{2}$$

As shown in the upper part of Fig. 2, the attention process compresses the feature map either vertically or horizontally and calculates self-attention separately for each axis. By incorporating horizontal and vertical axial attention modules, a global receptive field is achieved. However, this axial attention mechanism prioritizes essential spectral features but may overlook spatial local details. To address this, we incorporate an additional spatial feature extraction module comprising 3×3 convolution and batch normalization. This module supplements spatial details and preserves the sample's spatial structure.

III. Generalization Validation

In the deep learning model validation paradigm, the HS image is partitioned as the training dataset and the test dataset. The training dataset usually contains a small portion of pixels. For each pixel, its neighbor pixels (usually in a 7×7 patch) are also fed into the model to provide spatial and contextual information. The pixels in the training dataset are randomly selected from the entire HS image. It has been demonstrated that with only 5% pixels, deep learning models [20], [21], [22], [24], [25], [35], [36] can reach almost 100% classification accuracy on the test dataset, which was constructed by the rest 95% of the HS image pixels. However, the patches in the test dataset actually share many pixels with some patches in the training dataset. As shown in Fig. 3, the pixels and their neighbor pixels in the training dataset cover a large portion of the HS image although the training dataset contains only 5% of the HS image pixels. For an arbitrary sample in the test dataset, for example, the pixel in blue in Fig. 3, there is a large chance that the sample shares pixels with a sample in the training dataset. If the high classification accuracy is reached because of the shared pixels, or rather shared information, instead of the model generalization, the samples that share more pixels with the training dataset will have a higher probability of being classified correctly compared to those samples sharing less pixels with the training dataset. This phenomenon is observed in our following experiment.

We divide all the samples in the test dataset into three subsets according to their maximum overlap rates with samples in the training dataset. The three subsets are: 1) samples without overlap; 2) samples with an overlap rate between 0% and 50%; and 3) samples with an overlap rate higher than 50%. The classification accuracies are calculated for each subset on various popular models. As can be seen in Fig. 4, the test samples with a higher overlap rate exhibit significantly higher accuracy than samples with a lower overlap rate. Therefore, the high classification accuracy is reached because of the shared pixels instead of the model generalization.

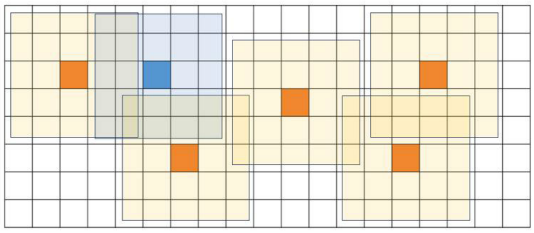


Fig. 3. Pixelwise random sampling dataset partition; 5% of the pixels are selected as the training dataset and marked in orange. The rest pixels (blue and white) are samples in the test dataset.

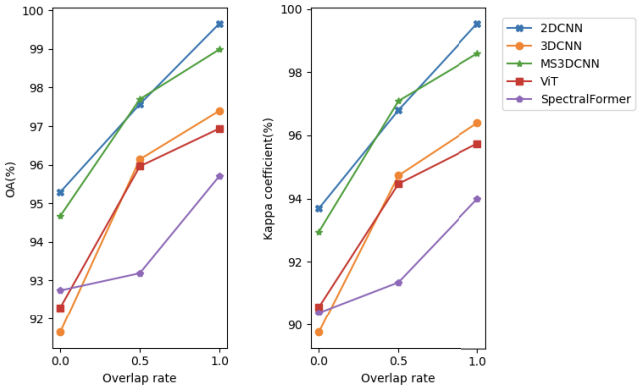


Fig. 4. Comparison of the accuracy of the overlap rate of training samples and test samples for different models.

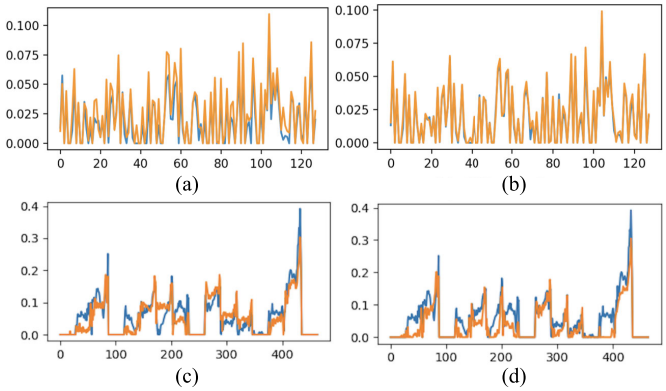


Fig. 5. Spectral features visualizations: (a) and (b) are spectral features extracted by the CNN model [22], while (c) and (d) are spectral features extracted by the 3DCNN model [25]. The blue and orange lines are spectral features of two samples from the same class, for instance (a) and (c), or different classes, such as (b) and (d).

As we discussed before, the samples in the training dataset and test dataset share information with each other if random sampling is applied. The shared information can lead to overly optimistic model performance. In this section, we introduce several dataset partition methods to provide a more comprehensive model generalization evaluation. If the model has good performance across different sampling methods, instead of merely random sampling, then we conclude that the model has good generalization capability. We further give a mathematical analysis of the information share for each dataset partition method.

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Furthermore, we visualize the spectral features extracted by 2DCNN [22] and 3DCNN [25] models for further analysis. The 2DCNN model achieved 99.35% accuracy, and the 3DCNN model achieved 97.36% accuracy on the PaviaU dataset. As shown in Fig. 5(a) and (b), for the 2DCNN model, the spectral patterns are very similar for the samples from the same class, as well as the samples from the different classes. For the 3DCNN model [see Fig. 3(c) and (d)], the spectral patterns are not very similar for the samples from both the same class and the different classes. However, for hyperspectral image analysis, the spectral information is the most important and should be distinguishable between classes.

Therefore, based on the above experiment and analysis, models trained under a random sampling paradigm can achieve good performance on the test dataset. However, the performance depends on the overlapping rate between the training samples and the test samples. Moreover, the spectral patterns extracted by the models cannot effectively differentiate between samples from different classes.

A. Dataset Partitions

1) Random Sampling: The most commonly used model validation paradigm for HS image classification is pixelwise random sampling. It randomly selects a small portion of pixels (5%) as the training dataset and leaves the rest of the pixels as the validation dataset. Each sample consists of a central pixel and its surrounding region (5×5 pixels patch) to provide spatial information. However, as we discussed above, random sampling can result in a high overlap between samples in the training dataset and the validation dataset, leading to a significant similarity between the training and validation datasets.

2) Checkerboard Sampling: To reduce the overlap between the samples in the training dataset and the samples in the validation dataset, we propose checkerboard sampling which partitions the training–test datasets through nonadjacent checkerboards. Compared to random sampling, samples that fall in the inner area of each checkerboard of the validation dataset will not overlap with the samples of the training dataset.

3) Blockwise Sampling: To further explore the generalization ability of the model, we propose blockwise sampling that divides the training and validation datasets into nonoverlapping horizontal or vertical blocks.

4) K-Means Sampling: In addition, we employ the k-means algorithm to cluster each class as k disjoint datasets and randomly select $k/2$ datasets as the training dataset while leaving the rest as the validation dataset.

The checkerboard, blockwise, and k-means samplings ensure that almost all the samples in the training dataset have no overlap with the samples in the validation dataset, while samples in the same class may gather in a small area in the entire HS image. The blockwise sampling can fail to separate all the classes equally. In this case, the k-means sampling can provide a better sample partition. For the model generalization validation, if the model has good performance across different sampling methods, instead of merely the random sampling, then we conclude that the model has proper generalization capability. Fig. 6 illustrates the aforementioned partitions.

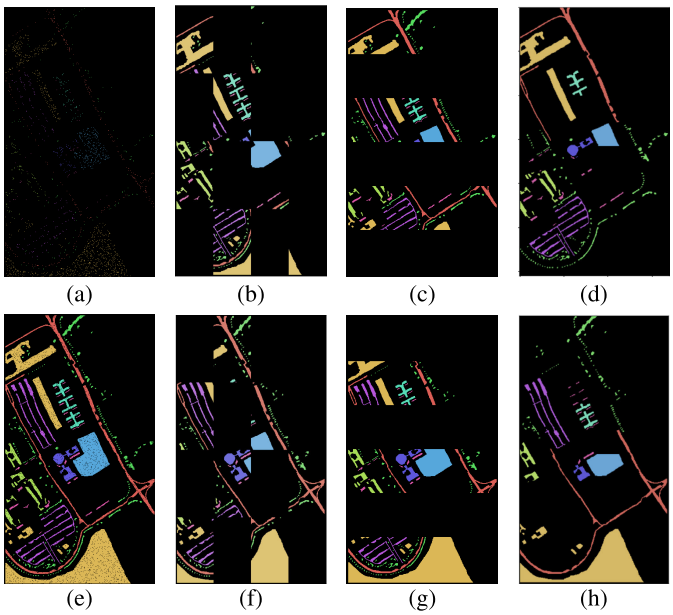


Fig. 6. Comparison diagram of the PaviaU dataset by four different methods to sample. (a) Trainset random. (b) Trainset checkerboard. (c) Trainset blockwise. (d) Trainset k-mean. (e) Testset random. (f) Testset checkerboard. (g) Testset blockwise. (h) Testset k-mean.

B. Information Share of Dataset Partitions

First, we interpret the classification using a set of randomly selected training samples as a random classification problem [66]. Specifically, we assume that Ω denotes the entire sample space, i.e., the hyperspectral image, and N is the total number of data samples. The sample space Ω is divided into mutually exclusive training dataset Ω_T and test dataset Ω_t through random permutation. Let α indicate the sampling ratio (SR) of the training dataset. Then, the sizes of Ω_T and Ω_t are $|\Omega_T| = \alpha N$ and $|\Omega_t| = (1-\alpha)N$, respectively, where $|\Omega| = N$. By the distribution of the random permutation, the probability that a sample $S(x, y)$ belongs to the training dataset or the test dataset is

$$P\{S \in \Omega_T\} = \alpha, \quad P\{S \in \Omega_t\} = 1 - \alpha. \tag{3}$$

The following result shows that the random sampling method causes a high probability of information shared between the training and test datasets, which could lead to an overestimated performance for the classifier.

Theorem 1: The dataset is divided into mutually exclusive training and testing sets through random permutation, where the training set accounts for α and the testing set accounts for $1-\alpha$. If the sample size is $n \times n$, for an arbitrary test sample S, the probability that it shares information with the training dataset is

$$p_I(S) = 1 - (1-\alpha)^{4n^2-1}. \tag{4}$$

The expected information benefit from the training dataset is

$$E[I(S)] = (n^2 - 1)\alpha H(S) \tag{5}$$

where $H(S)$ is the entropy of S.

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Proof: If any sample within the square centered at $S(x, y)$ of size $2n$ belongs to the training dataset, the two samples will have pixel overlap between each other, so as their information. Only if all the samples in the square belong to the test dataset, S does not share any information with the training dataset. Thus, by the independency of the samples, the probability that S shares information with the training dataset is

$$p_I(S) = 1 - P\{S \in \Omega_t\}^{(4n^2-1)}$$

where Ω_t is the test set. By the distribution of the random permutation, $P\{S \in \Omega_t\} = 1 - \alpha$. Thus,

$$p_I(S) = 1 - (1-\alpha)^{(4n^2-1)}.$$

As for the expected information benefit, we first define the information share ratio (ISR) between two samples $S(x, y)$ and $S'(x', y')$ as the ratio of the information that they share to the information of S as

$$\gamma(S, S') := \frac{I(S, S')}{H(S)} \tag{6}$$

where $I(S, S')$ is the mutual information. Assuming that each pixel contains the same amount of information, the ISR can be represented by the distance of the two samples as

$$\gamma(S, S') := \frac{I(S, S')}{H(S)} = \frac{i_x i_y}{n^2} \tag{7}$$

where $i_x = (n - |x - x'|)_+$, $i_y = (n - |y - y'|)_+$, and $(x)_+ = x\mathbf{1}_{x>0}$. The expected information benefit of S is

$$\begin{aligned} E[I(S)] &= \sum_{S'} E_{S'}[\gamma(S, S')H(S)] = \sum_{S' \in \Omega_T} \gamma(S, S')\alpha H(S) \\ &= \left(\left(\sum_{x'=x-n}^{x+n} \sum_{y'=y-n}^{y+n} \frac{i_x i_y}{n^2} \right) - 1 \right) \alpha H(S) \\ &= (n^2 - 1)\alpha H(S) \end{aligned} \tag{8}$$

where Ω_T is the training set. □

By Theorem 1, if we take a commonly used setting, i.e., dividing $\alpha = 5\%$ data as the training dataset and applying $n = 5$ pixels patch for each sample, the probability that an arbitrary sample in the test dataset shares information with the training dataset is $p_I(S) > 0.993$, and the expected information benefit is $1.24H(S)$. Thus, the samples in the test dataset can take a lot of information benefits through the sample overlapping, which contains even more information than the sample itself. Due to the information benefit, the information in the test dataset is already included in the training dataset. Hence, it is not surprising to see good performance on the test dataset. However, the classifier can fail to generalize on samples that it does not share or share less information with the training dataset.

Theorem 2: The dataset is divided through blockwise sampling, where the height of each block is h for rowwise (or the width of each block is w for columnwise). If the sample size is $n \times n$, for an arbitrary test sample S, the expected information benefit of the training dataset is

$$E(I(S)) = \frac{(n^2 - n)}{h} H(S). \tag{9}$$

Proof:

$$\begin{aligned} E[I(S)] &= \sum_{S'} E_{S'}[\gamma(S, S')H(S)] \\ &= \left(\left(\sum_{y'=y-n}^{y-1} \frac{i_y n}{n^2} + \sum_{y'=y+1}^{y+n} \frac{i_y n}{n^2} \right) \right) \frac{n}{h} H(S) \\ &= \frac{n^2 - n}{h} H(S). \end{aligned} \tag{10}$$

□

Proposition 1: Assume that the sample space Ω divided the training dataset Ω_T and the test dataset Ω_t through checkerboard sampling, i.e., nonadjacent checkerboards. If the sample size is $n \times n$ for an arbitrary test sample S and the height of each checkerboard is h and the width w, the expected information benefit for the checkerboard sampling is

$$E[I(S)] \leq (n^2 - n)\left(\frac{1}{h} + \frac{1}{w}\right) H(S). \tag{11}$$

Proposition 2: The expected information benefit for the k-means sampling is bounded by

$$\begin{aligned} E[I(S)] &\geq (n^2 - n)\left(\min\left(\frac{1}{h}, \frac{1}{w}\right)\right) H(S) \\ E[I(S)] &\leq (n^2 - n)\left(\frac{1}{h} + \frac{1}{w}\right) H(S) \end{aligned} \tag{12}$$

where h and w are the total height and width of the image divided by k.

Propositions 1 and 2 are intuitive to proof. Taking the University of Pavia dataset (610 × 340 pixels) as an example, if the dataset is divided into eight rows, the expected information benefit is $0.258H(S)$, which is about one-fifth of the information benefit of the 5% random sampling. If the 8×8 checkerboard sampling is taken, the expected information benefit is $0.519H(S)$. The expected information benefit of the k-means sampling can be upper bounded by checkerboard sampling and lower bounded by blockwise sampling. If $k = 4$, the expected information benefit is between $0.131H(S) \leq E[I(S)] \leq 0.365H(S)$.

From the specific values of the expected information benefit, it can be seen that the three proposed partition methods can make the test set less polluted by the training set than random sampling, so as to provide a better measure of the generalization ability and robustness of the classification model.

IV. Experiments

In this section, we first introduce several commonly used HS image datasets and the details for dataset partition methods. Then, we compare our proposed SaaFormer with other state-of-the-art methods under the various dataset partition methods to validate the generalization ability of the models. Extensive experiments are finally conducted, including ablation analysis, to validate the efficacy of each component in the proposed model.

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A. Data Description

Experiments are conducted on six hyperspectral datasets.[1] The detailed descriptions of these datasets are listed as follows.

1) The Indian Pines dataset consists of 145 × 145 pixels and 220 spectral bands. The dataset has 20 m per pixel spatial resolutions and 10-nm spectral resolutions covering a spectrum range of 200–2400 nm. In the experiment, after removing 20 noisy and water absorption bands, 200 spectral bands are retained. The ground truth (GT) is composed of 16 vegetation classes.
2) The Pavia Center dataset consists of 102 spectral bands, while the Pavia University dataset has 103 spectral bands. The Pavia Center image has a size of 1096 × 1096 pixels, and Pavia University is 610 × 610 pixels. Some samples within both images do not contain any information and need to be discarded for analysis purposes. Both datasets have GTs that are classified into nine different classes each.
3) The Salinas scene dataset was obtained using the 224-band AVIRIS sensor over Salinas Valley, California. It covers an area of 512 lines with 217 samples. The dataset also removed 20 bands affected by water absorption. The image was provided solely as at-sensor radiance data.
4) The Kennedy Space Center (KSC) dataset was collected from an altitude of around 20 km, providing a spatial resolution of 18 m. After eliminating bands affected by water absorption and low signal-to-noise ratio, the analysis was conducted using 176 bands. The dataset defined 13 classes that represent different land cover types.
5) The Botswana dataset was obtained at a pixel resolution of 30 m, covering a strip of 7.7 km in 242 bands. To enhance data quality, uncalibrated and noisy bands that corresponded to water absorption features were removed. The remaining 145 bands, comprising bands, were included as candidate features for analysis. The dataset consists of observations from 14 identified classes representing land cover types found in seasonal swamps, occasional swamps, and drier woodlands located in the distal portion of the delta.
6) The Houston2013 dataset was collected using the ITRES CASI-1500 sensor, covering the University of Houston campus and its surrounding rural areas in Texas, USA, which is widely recognized as a benchmark for evaluating land cover classification performance [44]. The hyperspectral (HS) cube consists of 349 × 1905 pixels with 144 wavelength bands ranging from 364 to 1046 nm at 10-nm intervals.

B. Data Partitions

1) Random Sampling: 5% of the samples from each class are randomly selected as the training set, and the remaining samples are used for testing. Each sample is $s \times s$ pixel patch.

[1]All these datasets can be obtained at https://www.ehu.eus/ccwintco/index.php/Hyperspectral_Remote_Sensing_Scenes.

TABLE I
SAMPLE NUMBERS FOR THE TRAINING AND TEST SETS FOR THE THREE DATA PARTITION METHODS

		Botswana	KSC	PaviaU	PaviaC	Salinas
Checkerboard	train	1403	2185	17153	57070	24993
	test	1845	3008	24395	89963	28353
Block-wise	train	1585	2043	14264	65076	25964
	test	1663	3168	28512	83076	28165
K-means	train	1170	1914	19982	67948	25517
	test	2078	3279	21566	79085	27829

2) Checkerboard Sampling: We divide the entire dataset into an equal number of blocks based on the height and width; for example, if we divide the whole dataset ($H \times W$) into $c \times c$ blocks, i.e., $[1, 2, \ldots, c \times c]$, each small block would have a size of $(H/c) \times (W/c)$. Then, using the interleaving method, we allocate 50% of the blocks as the training set and the remaining blocks as the test set. Specifically, let N_1 denote the total number of sample points in the odd sequence block and N_2 denote the total number of sample points in the even sequence block. If $N_1 < N_2$, then N_1 will be used as the training set; otherwise, N_2 will be used as the training set.

3) Blockwise Sampling: Taking a different approach from the checkerboard sampling, we opt to divide the dataset based on the shorter dimension. The dataset ($H > W$) is split into b portions, i.e., $[1, 2, \ldots, b]$; each portion would have a size of $H \times W/b$. We set b is the minimum value, which ensures that all classes are represented in both the training and testing sets. The selection of the training set is the same as before.

4) K-Means Sampling: Predefining the number of cluster centers k, we allocate all samples from different categories in the dataset to these k clusters using the k-means clustering method. Afterward, we select $k/2$ clusters as the training set and the remaining clusters as the test set. We also select a smaller sample area as the training set.

It is noteworthy that chessboard sampling techniques have already been applied in neural network sampling methods, as discussed in [65]. However, the main objective and application context of the sampling method are different from those of ours. Specifically, Arada and Dadios [65] introduced the checkerboard sampling aimed at addressing database memory issues. In contrast, our method focuses on addressing data leakage issues between training and test samples. Unlike the method in [65], which reduces the number of fingerprint templates in the database by performing checkerboard sampling on each sample image, our strategy involves directly applying checkerboard sampling across the whole dataset to effectively reduce the overlap between training and test samples.

Moreover, our proposed chessboard sampling differs significantly from block sampling in its handling of boundary samples. Checkerboard sampling discards boundary samples to prevent overlap between training and test samples, while blockwise sampling deals with the overlapping boundary regions with zero padding. By employing this differentiated sampling approach, we can more effectively validate the generalization capability and robustness of the proposed model. Due to significant differences in data distribution across various

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TABLE II
COMPARISON OF CLASSIFICATION ACCURACY OF DIFFERENT SAMPLING METHODS ON THE BOTSWANA DATASET

		1DCNN	2DCNN	3DCNN	MS3DCNN	RNN	ViT	SpectralFormer	MorphFormer	SSFTTnet	SaaFormer
Random	OA	0.8869	0.9818	0.9465	0.9345	0.8700	0.9102	0.9782	0.9559	0.9780	**0.9925**
	AA	0.8977	0.9757	0.9475	0.9306	0.8779	0.9136	0.9808	0.9440	0.9755	**0.9933**
	Kappa	0.8775	0.9803	0.9420	0.9290	0.8592	0.9028	0.9764	0.9522	0.9761	**0.9919**
Block	OA	0.8151	0.5968	0.7810	0.6529	0.7406	0.8296	0.6700	0.7811	0.8196	**0.9117**
	AA	0.8611	0.6169	0.7585	0.6786	0.8094	0.7870	0.6770	0.8266	0.8633	**0.9202**
	Kappa	0.7973	0.5645	0.7609	0.6233	0.7193	0.8141	0.6431	0.7619	0.8022	**0.9027**
Checkerboard	OA	0.8738	0.9505	0.8788	0.8392	0.7631	0.9181	0.9043	0.9766	0.8268	**0.9786**
	AA	0.9114	0.9569	0.9182	0.8879	0.8366	0.9301	0.9208	0.9713	0.8997	**0.9793**
	Kappa	0.8620	0.9455	0.8675	0.8252	0.7424	0.9101	0.8947	0.9742	0.8125	**0.9765**
K-means	OA	0.8639	0.8851	0.8905	0.8277	0.7158	0.8371	0.8921	0.8581	0.6350	**0.9122**
	AA	0.8365	0.8328	0.8611	0.7496	0.6672	0.8070	0.8405	0.8199	0.7533	**0.8661**
	Kappa	0.8510	0.8738	0.8799	0.8112	0.6879	0.8167	0.8817	0.8435	0.6088	**0.9037**

TABLE III
COMPARISON OF CLASSIFICATION ACCURACY OF DIFFERENT SAMPLING METHODS ON THE PAVIAU DATASET

		1DCNN	2DCNN	3DCNN	MS3DCNN	RNN	ViT	SpectralFormer	MorphFormer	SSFTTnet	SaaFormer
Random	OA	0.8820	0.9935	0.9121	0.9736	0.9057	0.9480	0.9625	0.9932	0.9947	**0.9980**
	AA	0.8619	0.9847	0.9130	0.9580	0.8833	0.9331	0.9499	0.9869	0.9914	**0.9958**
	Kappa	0.8416	0.9814	0.8825	0.9651	0.8743	0.9312	0.9503	0.9909	0.9930	**0.9973**
Block	OA	0.7868	0.7307	0.9391	0.9673	0.9223	0.9462	0.8992	0.8703	0.8309	**0.9747**
	AA	0.8405	0.7774	0.9086	0.9261	0.8774	0.8993	0.8188	0.8935	0.8163	**0.9695**
	Kappa	0.6934	0.6216	0.9026	0.9477	0.8767	0.9160	0.8405	0.8182	0.7671	**0.9634**
Checkerboard	OA	0.8908	0.9211	0.9011	0.9499	0.8102	0.8837	0.8775	0.9419	0.9483	**0.9666**
	AA	0.8970	0.8894	0.9055	0.9558	0.7894	0.8835	0.8992	0.9305	**0.9658**	0.9641
	Kappa	0.8510	0.8937	0.8665	0.9322	0.7412	0.8427	0.8361	0.9218	0.9309	**0.9543**
K-means	OA	0.8163	0.9071	0.8386	0.8403	0.7992	0.8498	0.8220	0.8498	0.8275	**0.9559**
	AA	0.8664	0.8892	0.8737	0.9259	0.8445	0.8841	0.8723	0.8810	0.8507	**0.9518**
	Kappa	0.7784	0.8861	0.8068	0.7884	0.7490	0.8192	0.7594	0.8200	0.7941	**0.9420**

datasets, Table I shows the sample numbers for the training and test sets for the three data partition methods.

C. Experimental Setting

The experiments will utilize various classification measures, as outlined in [66]. A summary of these measures is provided in the following.

1) C_m represents the set of data samples in the mth class as determined by the GT.
2) n_{mm} represents the number of samples in the mth class that is correctly classified into the mth class $\hat{C}_m$.
3) n_{jm} represents the number of data samples in the mth class C_m that is actually classified into the jth class $\hat{C}_j$.
4) $\hat{n}_{mj}$ represents the number of samples from the jth class that is classified as the mth class
5) M represents the number of classes.
6) $n_m = \sum_{j=1}^{M} n_{jm}$ denotes the number of data samples in the mth class C_m.
7) $\hat{n}_m = \sum_{m=1}^{M} \hat{n}_{mj}$ denotes the total number of data samples that are classified into mth class.
8) $N = \sum_{m=1}^{M} n_m$ represents the total number of data samples.
9) $\hat{N} = \sum_{m=1}^{M} \hat{n}_m$ denotes the total number of data samples that are classified.

1) Evaluation Metrics: In HSIC, commonly used performance measurements include class accuracy (CA), overall accuracy (OA), average accuracy (AA), and κ (Kappa), which can be expressed as

$$P_{\mathrm{OA}} = \frac{1}{N}\sum_{m=1}^{M} n_{mm} = \sum_{m=1}^{M} \frac{n_m}{N} p_A(C_m) \tag{13}$$

where $p_A(C_m) = (n_{mm}/\sum_{j=1}^{M} n_{jm}) = (n_{mm}/n_m)$ is the accuracy of the mth class. P_{OA} employs sample ratios as weights to calculate the AA of each class. In addition, we further consider all classes to have equal likelihood, i.e., P_{AA} as the AA

$$P_{\mathrm{AA}} = \frac{1}{M}\sum_{m=1}^{M} \frac{n_{mm}}{n_m} = \frac{1}{M}\sum_{m=1}^{M} p_A(C_m). \tag{14}$$

Moreover, the kappa coefficient is a measurement indicator used to assess the consistency between model predictions and true labels in deep learning classification tasks. It takes into account the possibility of random agreement in the predicted results and compares the model's predictions with random chance agreement

$$\kappa = \frac{P_{\mathrm{OA}} - p_e}{1 - p_e} = \frac{N\sum_{m=1}^{M} n_{mm} - \sum_{j=1}^{M} n_m \sum_{j=1}^{M} n_{jm}}{N^2 - \sum_{m=1}^{M} n_m \sum_{j=1}^{M} n_{jm}} \tag{15}$$

where $\kappa \in [-1, 1]$, 1 represents perfect agreement, 0 indicates that the predictions are as consistent as would be expected by chance, and -1 represents complete disagreement. A higher kappa coefficient indicates better classification performance of the model.

2) Comparison With State-of-the-Art Backbone Networks: To assess the classification performance of the proposed model, we compared it with some established methods for hyperspectral image classification (HSIC). These methods include 1DCNN classifier [19], 2DCNN-LR [22], 3-DCNN-based FE model [24], M3D-DCNN [25], and RNN [26], as well as four transformer-based models, the traditional ViT [35], SpectralFormer [36], morphFormer [37], and SSFTTnet [39]. The parameter configurations for these network structures and the training sample size are set according to the default settings specified in their respective references.

As for SaaFormer architecture, it employs an embedded spectrum with $d = 128$ channels. The embedding is further segmented into $K = 2$, $c_1 = 128$, and $c_2 = 32$ spectral clips,

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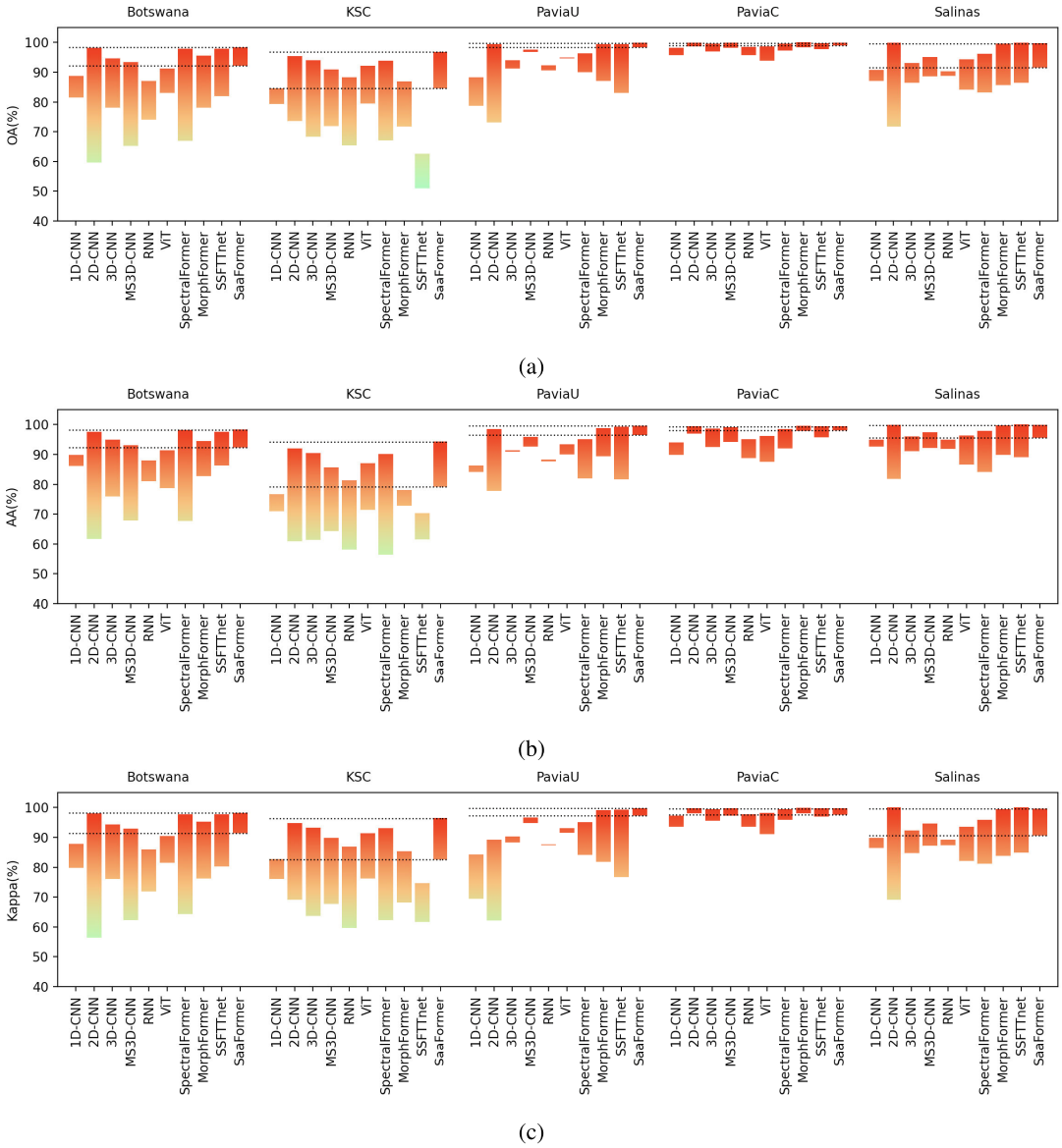


Fig. 7. Comparison of different models on classification accuracies in terms of OA, AA, and Kappa across five datasets. The top and bottom of the bars represent performance metrics under random sampling and blockwise sampling, respectively. (a) Overall Accuracy (OA), (b) Average Accuracy (AA), and (c) Kappa.

which are then processed through two consecutive transformer encoder blocks for classification. Each encoder block includes a four-head axial aggregation attention layer, an MLP layer, and a GELU nonlinear activation layer. To enhance model robustness, a dropout layer is applied after encoding positional embeddings and within the MLPs, with a 40% dropout rate. We use the Adam optimizer [67] with a batch size of 128 to train the model with a weight decay of 5e-3 and a learning rate of 5e-4. In addition, these methods used a step scheduler with a gamma of 0.9, steps of size tenth of the total epochs, and trained during about 500 epochs.

D. Generalization Validation

To assess the generalization capabilities of the proposed model and several popular models, experiments are conducted on the mentioned datasets. The evaluation is based on classification results, using metrics OA, AA, and κ.

The visualization illustrates the changes in the accuracy of the classification model across various sampling strategies using bar charts. The length of each bar indicates the magnitude of accuracy variation, with longer bars indicating more significant fluctuations and shorter bars suggesting more stable performance. Specifically, in Fig. 7, the top and bottom of the bars correspond to the model's performance metrics when employing random sampling and blockwise sampling, respectively. On the other hand, in Fig. 8, the top and bottom of the bars represent the model's metrics when applying k-means sampling and checkerboard sampling.

The results from Figs. 7 and 8 underscore the high accuracy of classification results obtained via random sampling for partitioning. This further corroborates the presence of data leakage in such partitioning schemes, leading to overfitting of the models. There exist significant disparities among different datasets, resulting in varied performance

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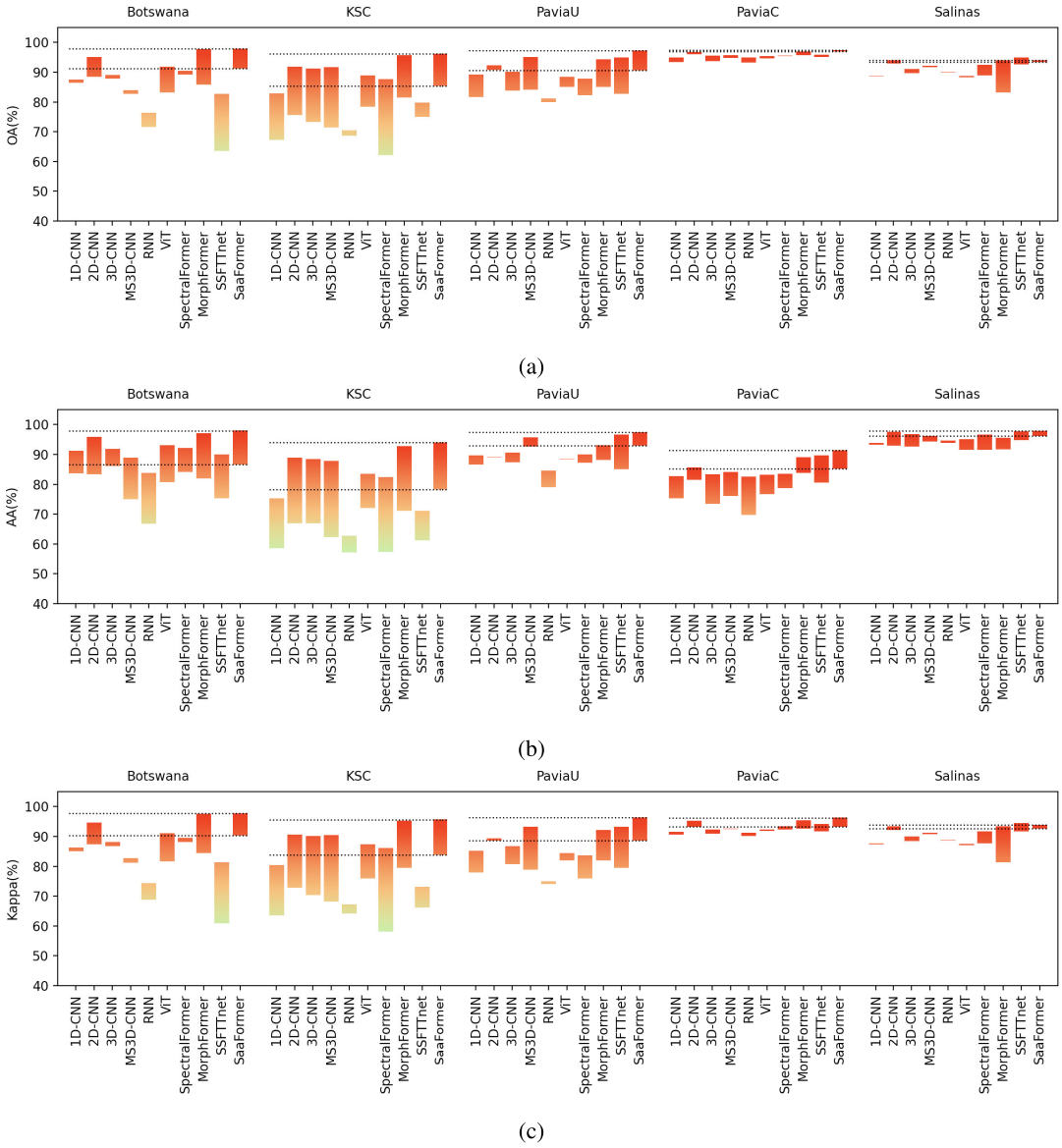


Fig. 8. Comparison of different models on classification accuracies in terms of OA, AA, and Kappa across five datasets. The top and bottom of the bars represent performance metrics under checkerboard sampling and k-means sampling, respectively. (a) Overall Accuracy (OA), (b) average Accuracy (AA), and (c) Kappa.

across models. Botswana and KSC datasets, characterized by their limited data volume and sparse distribution, exhibit considerable fluctuations in model accuracies. Notably, the 2DCNN models are more susceptible to spatial distribution effects, whereas 1DCNN models emphasize spectral features, thus showing relatively smoother fluctuations. In addition, the poorer performance of SSFTTnet on the KSC dataset may be attributed to the adverse impact of PCA on spectral information. In contrast, the latter three datasets, with larger volumes and denser distributions, demonstrate overall less variability. Under random sampling, MorphFormer, SSFTTNet, and SaaFormer achieve remarkably high accuracies, albeit with significant fluctuations when partitioning methods are altered. In comparison, our model exhibits greater stability, maintaining optimal performance even under alternative partitioning strategies. In summary, these visualizations eloquently highlight the superiority of our model, which not only excels in accuracy but also demonstrates exceptional generalization capabilities and robustness.

Tables II and III present detailed results regarding the three evaluation metrics on the Botswana and PaviaU datasets. The best accuracy is highlighted in bold. Our model consistently achieves the highest accuracy in nearly all cases. Moreover, it is noteworthy that the variations in the three evaluation metrics among different partitioning methods are minimal, typically within a 1% range. This highlights that our model consistently delivers high accuracy across diverse datasets and remains stable under various partitioning methods, underscoring its exceptional generalization and robustness.

E. Model Analysis

1) Ablation Study: We first conducted experiments to determine the optimal length for spectral clips. With $c_1 = 128$ and $K = 2$ fixed, we evaluated spectral clips of lengths 8, 16, 32, and 64 for c_2. Experiments were performed on four datasets, and the specific experimental results are shown in Table IV.

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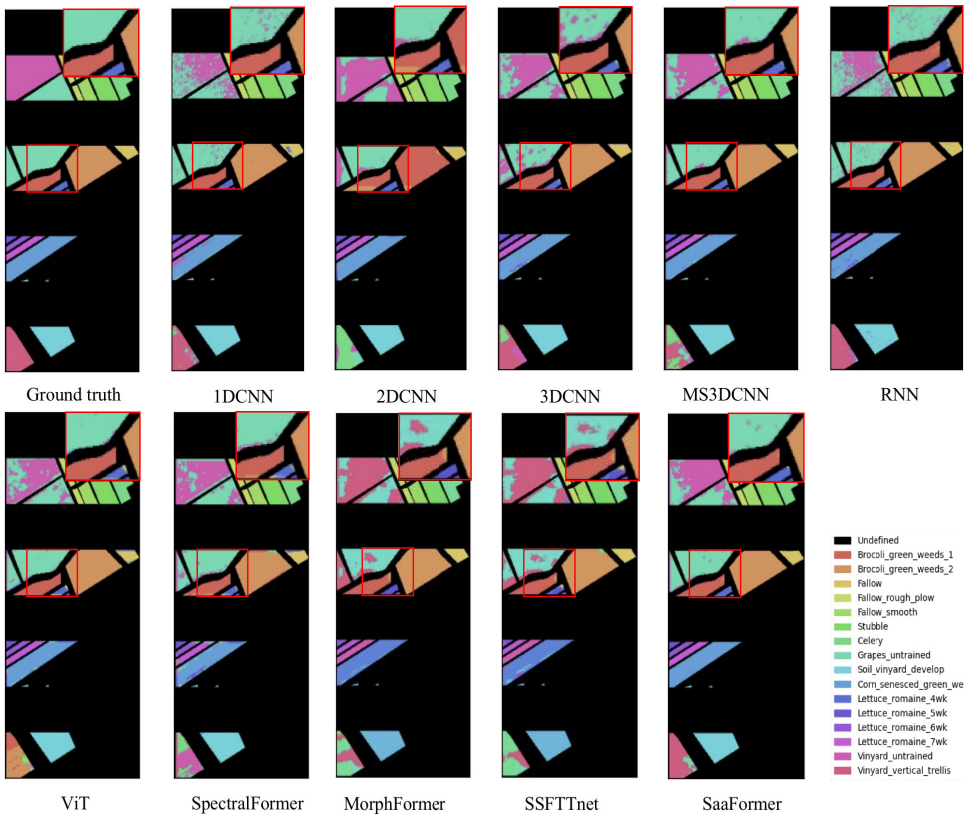


Fig. 9. Spatial distribution of testing sets using blockwise sampling and the classification maps obtained by different models on the Salinas dataset.

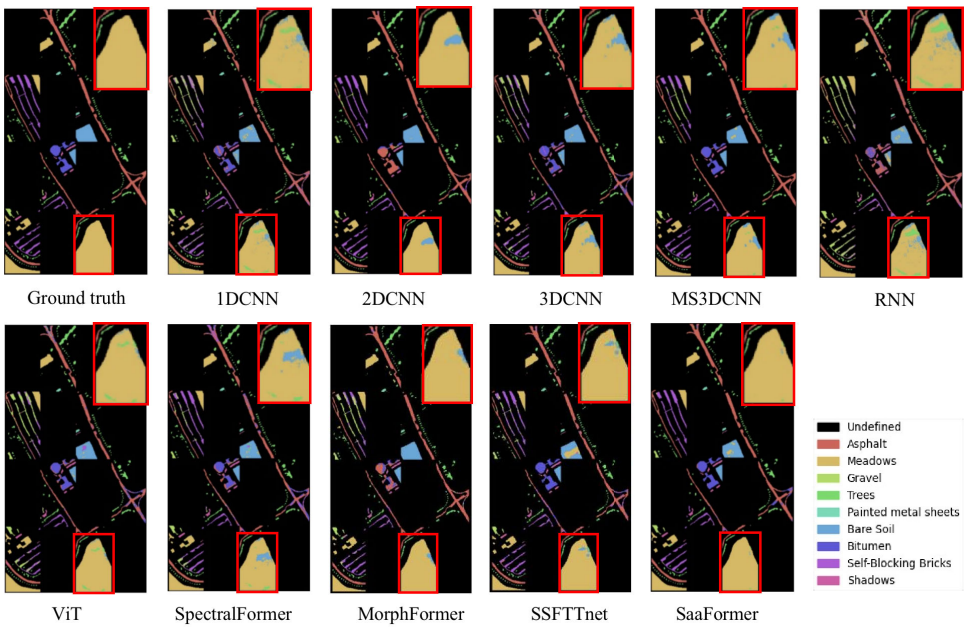


Fig. 10. Spatial distribution of testing sets using random sampling and the classification maps obtained by different models on the PaviaU dataset.

We further investigate the optimal number K of the multilevel spectral extraction. For each level, we preset $c_1 = 128$, $c_2 = 64$, $c_3 = 32$, and $c_4 = 16$ to extract different granular spectral features. For different datasets, we set K according to the optimal spectral clips in Table V. The results indicate that our model exhibits higher accuracy when incorporating multilevel spectral extraction components ($K = 2, 3$) compared to relying solely on the complete spectral information ($K = 1$). It indicates the effectiveness of the multilevel spectral extraction structure in capturing underlying patterns and correlations within the spectral data.

2) Computational Complexity: For a given HS image, let $X \in \mathbb{R}^{h \times w \times c}$ be the input sample. The per-layer time complexity of the proposed SaaFormer is dominated by axial

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TABLE IV

CLASSIFICATION PERFORMANCE ANALYSIS OF DIFFERENT SPECTRAL CLIP LENGTHS ACROSS VARIOUS DATASETS. WITH $c_1 = 128$ FIXED, SPECTRAL CLIPS OF LENGTHS 8, 16, 32, AND 64 FOR c_2 WERE EVALUATED

		random				block-wise				k-means				chessboard			
		$c_2 = 8$	16	32	64	$c_2 = 8$	16	32	64	$c_2 = 8$	16	32	64	$c_2 = 8$	16	32	64
Botswana	OA	0.983	**0.993**	0.989	0.987	0.909	**0.912**	0.890	0.892	0.838	0.874	**0.888**	0.855	0.921	0.933	0.939	**0.963**
	AA	0.985	**0.993**	0.991	0.989	0.913	**0.920**	0.900	0.892	0.876	0.884	**0.893**	0.825	0.944	0.954	0.938	**0.940**
	Kappa	0.981	**0.992**	0.988	0.986	0.900	**0.903**	0.879	0.882	0.819	0.860	**0.875**	0.837	0.913	0.926	0.933	**0.959**
KSC	OA	0.904	0.918	**0.930**	0.929	0.842	0.846	**0.867**	0.838	**0.811**	0.808	0.794	0.781	0.960	0.956	**0.964**	0.954
	AA	0.834	0.848	**0.873**	0.883	0.814	0.815	**0.818**	0.805	**0.738**	0.737	0.711	0.691	0.933	**0.939**	0.936	0.927
	Kappa	0.893	0.909	**0.922**	0.920	0.825	0.825	**0.848**	0.816	**0.791**	0.787	0.771	0.757	0.955	0.950	**0.959**	0.95
PaviaU	OA	0.991	0.992	**0.992**	0.992	0.972	0.972	**0.982**	0.970	0.923	0.918	**0.956**	0.926	0.958	0.959	**0.967**	0.965
	AA	0.987	0.989	**0.988**	0.988	0.968	**0.968**	0.967	0.969	0.942	0.940	**0.952**	0.934	0.964	0.963	**0.964**	0.971
	Kappa	0.989	0.989	**0.990**	0.989	0.959	0.959	**0.972**	0.956	0.901	0.894	**0.942**	0.903	0.943	0.944	**0.954**	0.953
Salinas	OA	0.965	0.966	**0.975**	0.971	0.908	0.908	**0.925**	0.900	0.888	0.887	**0.895**	0.892	0.917	0.918	**0.922**	0.917
	AA	0.985	0.981	**0.988**	0.986	0.954	0.953	**0.964**	0.954	0.949	0.949	**0.950**	0.948	0.947	0.947	**0.949**	0.946
	Kappa	0.961	0.962	**0.972**	0.967	0.896	0.897	**0.915**	0.888	0.875	0.874	**0.883**	0.878	0.908	0.909	**0.914**	0.908

TABLE V

ABLATION ANALYSIS OF THE PROPOSED SAAFORMER WITH A COMBINATION OF DIFFERENT STRUCTURES ON DIFFERENT DATASETS

		random			block-wise			k-means			chressborad		
	K	OA	AA	kappa	OA	AA	kappa	OA	AA	kappa	OA	AA	kappa
Botswana	1	0.9708	0.9744	0.9684	0.9041	0.9184	0.8947	0.8154	0.8330	0.7944	0.8925	0.9398	0.8829
	2	**0.9925**	**0.9933**	**0.9919**	0.9117	0.9202	0.9027	**0.9122**	**0.8661**	**0.9037**	**0.9631**	**0.9400**	**0.9594**
	3	0.9752	0.9747	0.9724	0.9098	0.9168	0.9009	0.8726	0.8722	0.8584	0.9012	0.9386	0.8924
	4	0.9646	0.9644	0.9617	**0.9135**	**0.9204**	**0.9050**	0.8667	0.8709	0.8508	0.9213	0.9346	0.9137
KSC	1	0.9048	0.8424	0.8939	0.8350	0.8123	0.8128	0.7748	0.7049	0.7443	0.9432	0.9205	0.9355
	2	**0.9303**	**0.8730**	**0.9223**	**0.8668**	**0.8176**	**0.8475**	**0.8114**	**0.7383**	**0.7907**	**0.9638**	**0.9357**	**0.9588**
	3	0.9125	0.8629	0.9026	0.8556	0.8153	0.8354	0.7973	0.7154	0.7747	0.9579	0.9301	0.9521
	4	0.9107	0.8561	0.9006	0.8218	0.7344	0.7968	0.7591	0.6582	0.7323	0.9510	0.9316	0.9433
PaviaU	1	0.9898	0.9843	0.9865	0.9750	0.9654	0.9637	0.8898	0.9359	0.8594	0.9063	0.9259	0.8722
	2	**0.9921**	**0.9881**	**0.9895**	**0.9824**	0.9673	**0.9720**	**0.9559**	**0.9518**	**0.9420**	**0.9666**	**0.9641**	**0.9543**
	3	0.9877	0.9791	0.9837	0.9764	**0.9684**	0.9659	0.9133	0.9262	0.8870	0.9144	0.8910	0.8831
	4	0.9910	0.9858	0.9881	0.9682	0.9663	0.9536	0.8924	0.9181	0.8611	0.9066	0.9272	0.8738
PaviaC	1	0.9948	0.9815	0.9927	0.9878	0.9767	0.9837	0.9695	0.9052	0.9557	0.9599	0.8511	0.9324
	2	**0.9976**	**0.9915**	**0.9966**	**0.9908**	**0.9795**	**0.9877**	**0.9713**	**0.9180**	**0.9583**	0.9559	0.8404	0.9257
	3	0.9956	0.9862	0.9938	0.9889	0.9768	0.9851	0.9695	0.9091	0.9556	**0.9610**	**0.8593**	**0.9344**
	4	0.9959	0.9845	0.9942	0.9834	0.9757	0.9770	0.9685	0.9055	0.9542	0.9586	0.8535	0.9303
Salinas	1	0.9664	0.9863	0.9626	0.8847	0.9419	0.8704	0.8893	0.9481	0.8763	0.9116	0.9402	0.9019
	2	**0.9749**	**0.9878**	**0.9721**	**0.9246**	**0.9637**	**0.9154**	0.8951	0.9503	0.8827	**0.9222**	**0.9490**	**0.9136**
	3	0.9734	0.9874	0.9703	0.9088	0.9525	0.8975	**0.8963**	**0.9517**	**0.8837**	0.9155	0.9395	0.9060
	4	0.9714	0.9866	0.9681	0.9030	0.9501	0.8911	0.8871	0.9428	0.8739	0.9147	0.9419	0.9050

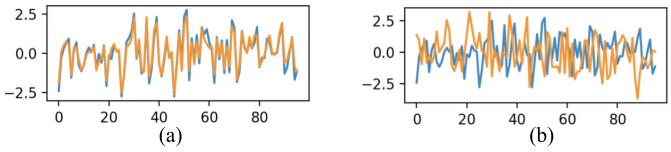


Fig. 11. Visualization of the spectral curves of samples feature maps after passing through the axial transformer block on the PaviaU dataset. The orange and blue curves in the figure represent the spectral characteristic plots of different sample points in the test set. The blue curves in both (a) and (b) correspond to the same test sample. In (a), the orange curve represents the spectral characteristic plot of a sample belonging to the same category as the blue curve, while, in (b), the orange curve represents the spectral characteristic plot of a sample from a different category than the blue curve.

TABLE VI

COMPUTATIONAL COMPLEXITY STUDIES ON COMPONENTS IN MULTILEVEL SPECTRAL EXTRACTION ON FOUR DATASETS (PAVIAU AND PAVIAC DATASETS HAVE THE SAME COMPUTATIONAL COMPLEXITY)

		$K = 1$	$K = 2$	$K = 3$	$K = 4$
Botswana	FLOPs	25.87M	828.22M	1218.81M	1442.13M
	Params	1.57M	4.56M	6.09M	7.83M
KSC	FLOPs	25.97M	828.32M	1218.90M	1442.23M
	Params	1.58M	4.56M	6.09M	7.83M
PaviaU	FLOPs	25.74M	828.09M	1218.67M	1442.00M
	Params	1.57M	4.55M	6.08M	7.82M
Salinas	FLOPs	26.06M	828.41M	1218.99M	1442.32M
	Params	1.58M	4.56M	6.10M	7.83M

self-attention and the spatial-enhanced modules. When a traditional attention module is applied on a feature map of x, the time complexity can be $\mathcal{O}(3h^2w^2d)$. However, for the axial aggregation attention, the time complexity is $\mathcal{O}(3(h + w)d + 2(h^2 + w^2)d)$. Thus, our axial aggregation attention can successfully reduce time complexity. The spatial-enhanced modules take $\mathcal{O}(k^2hwd^2)$, where k is the size of the convolution kernel and d is the size of the hidden features. Furthermore, we compared the computational complexities of models with different spectral extraction levels K in Table VI.

3) Spectral Variability: Hyperspectral imagery collected from airborne or satellite sources inevitably suffers from

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spectral variability, making it difficult for spectral classification [68]. Spectral variability refers to a variation of a spectral signature for a given material due to illumination conditions and topography, atmospheric effects, or even the intrinsic variability of the material. We conducted a comparative analysis to evaluate the performance of different models when faced with spectral variability on the Botswana dataset. According to [68], a 25-dB white Gaussian noise was added to the dataset and subsequently reevaluated the performance of different models using various partitioning methods. Detailed information regarding the experimental outcomes can be found in Table VII.

Based on the findings presented in Table VII, it is evident that the introduction of noise interference leads to a varying degree of degradation in the classification performance of each model. Nonetheless, our proposed model consistently attains the highest level of accuracy.

F. Visualization

1) Classification Result Visualization: We perform a qualitative assessment by visually representing the classification maps produced by various methods. Fig. 9 shows the results of the Salinas dataset, while, respectively, Fig. 10 presents the results on the PaviaU dataset using different dataset division methods.

By visualizing the datasets, it is evident that our model produces highly satisfactory classification maps, particularly with regard to texture and edge details. Within the region of interest (ROI), our model displays notably more realistic and intricate details. Consequently, the visual representation of the classification results unequivocally showcases that our model excels in generating highly favorable classification maps, especially in terms of texture and edge details.

2) Feature Extraction Visualization: Furthermore, when employing blockwise sampling, we randomly select a set of samples from the PaviaU dataset, both from the same class and different classes. We visualize the spectral pattern of their feature maps after passing through the axial transformer block. Fig. 11(a) represents the spectral pattern of samples from the same class, while Fig. 11(b) represents the spectral pattern of samples from different classes. Samples of the same class exhibit highly similar spectral patterns, while those from different classes show significant differences. This further demonstrates that our model excels at distinguishing between different types of samples, which yields more accurate classification performance, displaying enhanced generalization and robustness capabilities.

3) Spectral Feature Visualization: We make a qualitative evaluation by visualizing the spectral feature map obtained by the proposed SaaFormer framework on the Botswana dataset. According to the results of the above ablation experiment, Fig. 12 visualizes selected spectral encoder output feature maps with $K = 2$. We randomly selected a sample from the test set and set the number of axial aggregation attention layers in our model to $n = 2$. We visualized the feature maps obtained from the last two layers. Specifically, for a spectral feature map $\mathbf{x} \in \mathbb{R}^{h\times w\times c_i}$, we reshaped it to $\mathbf{x} \in \mathbb{R}^{hw\times c_i}$,(i = 1,2), where the color intensity represents the relative magnitude of the spectral values. The results demonstrate that a longer spectral clip ($c_1 = 128$) leads to lesser variation in the amplitude of the spectral feature map, highlighting a greater emphasis on extracting overall features. Conversely, a shorter spectral clip ($c_2 = 32$) results in deeper colors and larger amplitudes, signifying a heightened focus on extracting detailed features. This also demonstrates the effectiveness and superiority of the designed multilevel spectral extraction module from the visual perspective. Furthermore, in order to provide a more comprehensive visual inspection of test methods, Figs. 13–15 show their classification maps for the other three datasets, i.e., Botswana, KSC, and PaviaC datasets using 5% of training samples.

G. Background Classification

As mentioned in [66] and [69], HSIC analysis has been conducted based on the provided GT, with the exclusion of background (BKG) samples from the classification process. A data sample is classified as a background (BKG) sample simply because it cannot be labeled. Two main reasons lead to the presence of BKG samples. First, the complexity of the data being analyzed may make it prohibitively expensive to assign complete GT labels. Second, hyperspectral imaging sensors have the capability to detect numerous subtle material substances that may not be identifiable through existing knowledge or visual inspection methods. Following [66], we choose P_{PR}, P_{OPR} and P_{APR} as metrics of BKG

$$p_{\mathrm{PR}}(C_m) = \frac{\hat{n}_{mm}}{\hat{n}_m} \tag{16}$$

$$P_{\mathrm{OPR}} = \sum_{m=1}^{M} p(\hat{C}_m) p_{\mathrm{PR}}(\hat{C}_m) = \sum_{m=1}^{M} \frac{\hat{n}_m}{\hat{N}} p_{\mathrm{PR}}(C_m) \tag{17}$$

$$P_{\mathrm{APR}} = \frac{1}{M} \sum_{m=1}^{M} p_{\mathrm{PR}}(\hat{C}_m). \tag{18}$$

We conducted experimental comparisons on three datasets [66]. Instead of being treated as a separate class, background samples are considered to potentially belong to a known class that cannot be labeled. During training, only samples with GT labels are used, while, during testing, the background samples are included. For the Indiana Pines data, the number of training samples is 102 (SR = 1%) with the number of selected training sample numbers for each class tabulated in Table VIII. Unlike the Indian Pines data, Salinas data have a large number of data samples. We selected small training sizes with 0.1% of total data samples as training samples. This ensured that the training sample sizes for each class were roughly equal, ranging from 3 to 4, respectively, according to the training class sizes allocated in Table IX. For the University of Pavia data, the training sample size for each class is roughly set ranging from 4 to 7, as listed in Table X, with the total number of 428 training samples with approximate SR = 0.1%, which was the same as that used for Salinas. The above SR selections follow [66] exactly and train in an iterative way. Then, we evaluate the performance metrics of our model in comparison to IRTS-3D-CNN [66].

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TABLE VII

COMPARATIVE PERFORMANCE ANALYSIS OF MODELS UNDER SPECTRAL VARIABILITY ON THE BOTSWANA DATASET

		1DCNN	2DCNN	3DCNN	MS3DCNN	RNN	ViT	SpectralFormer	MorphFormer	SSFTTnet	SaaFormer
Random	OA	0.7547	0.9588	0.8931	0.6056	0.5629	0.7926	0.8837	0.9543	0.9064	**0.9802**
	AA	0.7622	0.9584	0.8947	0.5805	0.5425	0.7893	0.8893	0.9449	0.8889	**0.9794**
	Kappa	0.7341	0.9554	0.8841	0.5718	0.5258	0.7752	0.8740	0.9505	0.8984	**0.9786**
Block	OA	0.5584	0.5905	0.7129	0.6719	0.5445	0.6890	0.6833	0.7476	0.7451	**0.8972**
	AA	0.6898	0.5922	0.7071	0.6686	0.6685	0.7436	0.6776	0.7281	0.7090	**0.9054**
	Kappa	0.5276	0.5558	0.6890	0.6447	0.5122	0.6630	0.6580	0.7265	0.7213	**0.8871**
Checkerboard	OA	0.6125	0.9453	0.8187	0.8079	0.6381	0.8031	0.9074	0.9406	0.8382	**0.9731**
	AA	0.7079	0.9544	0.8778	0.8698	0.7355	0.8686	0.9325	0.9500	0.8817	**0.9788**
	Kappa	0.5811	0.9399	0.8029	0.7914	0.6090	0.7867	0.8984	0.9347	0.8233	**0.9704**
K-means	OA	0.7120	0.8901	0.8863	0.8545	0.6479	0.7906	0.8853	0.9184	0.5816	**0.9436**
	AA	0.7164	0.8424	0.8438	0.8407	0.6518	0.8347	0.8163	0.8694	0.6740	**0.9331**
	Kappa	0.6839	0.8781	0.8740	0.8404	0.6115	0.7698	0.8728	0.9098	0.5486	**0.9378**

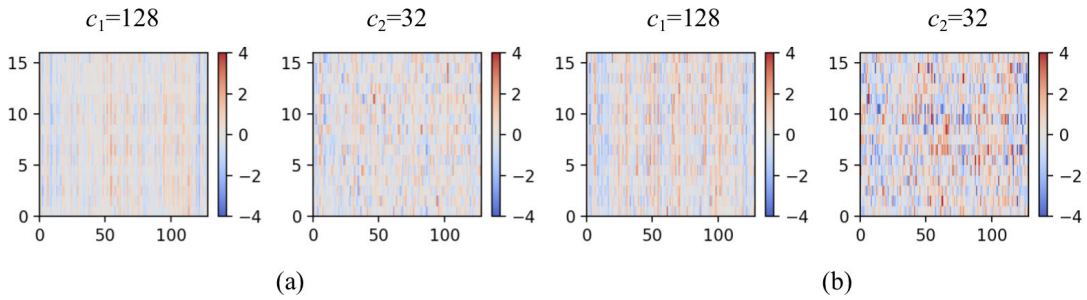


Fig. 12. Visualization of selected spectral encoder output feature maps obtained with different lengths of spectral channels ($K = 2, c_1 = 128$, and $c_2 = 32$). We randomly selected a sample from the test set and set the number of axial aggregation attention layers in our model to $n = 2$ and visualized the feature maps $\mathbf{x} \in \mathbb{R}^{hw \times c_i}$, $(i = 1, 2)$ obtained from the two layers, where the color intensity represents the relative magnitude of the spectral values. (a) Visualization of the spectral feature maps with spectral lengths of $c_1 = 128$ and $c_2 = 32$ from the first layer feature outputs, while (b) displays the visualization of the second layer feature outputs.

TABLE VIII

NUMBER OF TRAINING SAMPLES ALLOCATED TO EACH CLASS FOR INDIANA PINES WITH NUMBER OF DATA SAMPLES IN EACH CLASS

Class	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Total
number of training sample	6	6	6	6	7	7	6	6	6	6	7	7	6	7	7	6	102
total number of each class	46	1428	830	237	483	730	28	478	20	972	2455	593	205	1265	386	93	10249

TABLE IX

NUMBER OF TRAINING SAMPLES ALLOCATED TO EACH CLASS FOR SALINAS WITH NUMBER OF DATA SAMPLES IN EACH CLASS

Class	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Total
number of training sample	3	3	3	3	4	4	3	4	4	3	3	3	3	3	4	4	54
total number of each class	2009	3726	1976	1394	2678	3959	3579	11271	6203	3278	1068	1927	916	1070	7268	1807	54129

TABLE X

NUMBER OF TRAINING SAMPLES ALLOCATED TO EACH CLASS FOR PAVIAU WITH NUMBER OF DATA SAMPLES IN EACH CLASS

Class	1	2	3	4	5	6	7	8	9	Total
number of training sample	4	6	5	7	4	4	4	5	4	43
total number of each class	6631	18649	2099	3064	1345	5029	1330	3682	947	42776

We conducted research on two experimental conditions concerning the Indian Pines dataset, primarily focusing on the inclusion of background samples during the training process. N_{bk} represents the number of background samples in the training dataset. To comprehensively evaluate model performance, we used different numbers of background samples. Initially, we set $N_{\text{bk}} = 0$, indicating the absence of any background samples in the training dataset. Subsequently, we set $N_{\text{bk}} = 7$ to ensure an equal number of background samples compared to other classes in the training dataset. In addition, we set $N_{\text{bk}} = 107$ (with a total of 10 776 background samples) to maintain a sampling rate of 1% for background samples. Furthermore, we explored cases where N_{bk} assumes values between and above these specified values to further observe fluctuations in model accuracy.

The performance comparisons on the Indian Pines dataset in Table XI. It is evident that SaaFormer demonstrates superior classification performance compared to IRTS-3D-CNN. When considering the inclusion of background (BKG) in the classification process to compute P_{OPR}(BKG) and P_{APR}(BKG), SaaFormer still outperforms RTS-3D-CNN. Furthermore, Figs. 16–19 show their classification maps without/with BKG for the three datasets, respectively.

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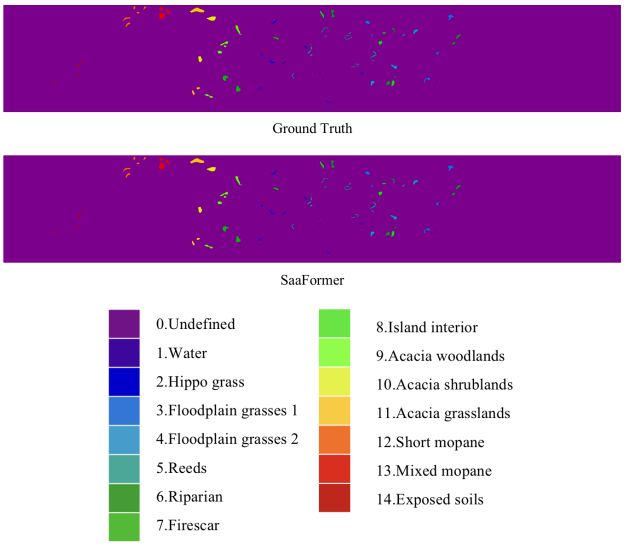


Fig. 13. Classification maps of SaaFormer for the Botswana dataset using 5% of training samples.

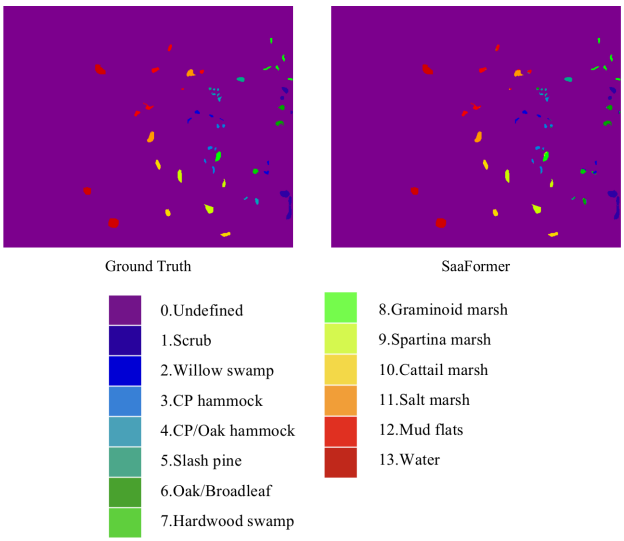


Fig. 14. Classification maps of SaaFormer for the KSC dataset using 5% of training samples.

In response to the background issue, we conducted further analysis regarding the impact of varying quantities of background samples on precision P_{PR}. It is discerned from the data presented in Table XII that augmenting the number of background samples N_{bk} in the training set significantly influences P_{PR}. As the number of background samples increases, both P_{OPR}(BKG) and P_{APR}(BKG) of SaaFormer notably surpass those of IRTS-3D-CNN. Particularly noteworthy is that at a threshold of 27 background samples, SaaFormer demonstrates markedly elevated precision, with subsequent increments in background sample count yielding marginal changes in model precision. If we looked at classes 2 and 15, our model was significantly improved after it was introduced, and the additional background samples resulted in pronounced enhancements in precision compared to IRTS-3D-CNN. The performance comparisons on the Salinas dataset in Table XIII and the PaviaU dataset in Table XIV. In order to provide a visual

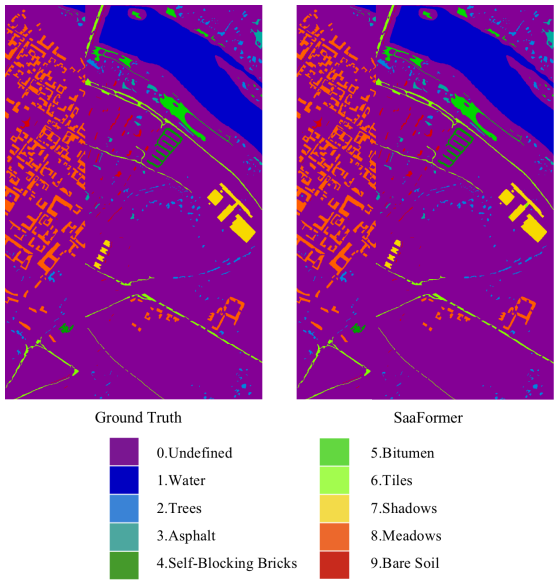


Fig. 15. Classification maps of SaaFormer for the PaviaC dataset using 5% of training samples.

TABLE XI

P_{PR}, P_{OA}, AND P_{AA} OF IRTS-3D-CNN AND SAAFORMER BY USING 1% TRAINING SAMPLES AND $N_{\mathrm{bk}} = 0$ FOR THE INDIAN PINES DATASET WITH THE HIGHEST ACCURACY BOLDFACED

	IRTS-3D-CNN		SaaFormer	
Class	$P_A(C_m)$	$P_{PR}(C_m)$	$P_A(C_m)$	$P_{PR}(C_m)$
1(alfalfa)	99.13	**34.64**	**100**	30.08
2(Corn_N)	78.60	**60.34**	**89.31**	54.60
3(Corn_M)	**91.64**	59.85	76.82	**69.27**
4(Corn)	97.81	**56.55**	**98.70**	53.60
5(Grass_P)	92.34	**35.26**	**95.59**	23.62
6(Grass_T)	98.67	34.50	**99.86**	**46.60**
7(Grass_M)	100	**26.13**	**100**	7.86
8(Hay_W)	99.64	**71.88**	**100**	67.24
9(Oats)	**100**	21.43	**100**	**25.00**
10(Soybean_N)	**87.70**	56.40	83.64	**57.73**
11(Soybean_M)	**77.84**	76.95	77.28	**84.85**
12(Soybean_C)	90.35	49.53	**96.08**	**53.45**
13(Wheats)	**100**	65.13	**100**	**65.25**
14(Woods)	97.91	**26.76**	**98.41**	26.38
15(Building)	98.08	16.37	**100**	**20.07**
16(Tower)	99.79	**42.23**	**100**	21.75
$P_{OA}(\%)$	88.45		**88.91**	
$P_{AA}(\%)$	94.34		**94.73**	
$P_{OPR}(BKG)(\%)$	43.12		**45.21**	
$P_{APR}(BKG)(\%)$	**45.87**		44.21	
Time(s)	2448.4		285.2	

inspection of test methods, Fig. 18 shows their classification maps using different numbers of background samples for the Indian Pines dataset, respectively. This underscores the potency of incorporating background samples in bolstering the model discriminative capacity and enhancing its generalization capabilities.

Python and PyTorch libraries were used to run the experiments. All experimental results are generated on a desktop with 16 vCPU Intel[2] Xeon[2] Platinum 8352V CPU and NVIDIA GeForce RTX 4090. In terms of parameter setting, we used a patch size of 7 and trained the models for 1000 epochs with a batch size of 128, and the embedding

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TABLE XII

P_{PR} OF IRTS-3D-CNN AND SAAFORMER USING DIFFERENT N_{bk} FOR THE INDIAN PINES DATASET

	IRTS-3D-CNN	SaaFormer				
		$N_{bk}=7$	$N_{bk}=27$	$N_{bk}=67$	$N_{bk}=107$	$N_{bk}=207$
Class	$P_{PR}(C_m)$	$P_{PR}(C_m)$	$P_{PR}(C_m)$	$P_{PR}(C_m)$	$P_{PR}(C_m)$	$P_{PR}(C_m)$
1(alfalfa)	34.64	43.33	56.34	40.40	79.07	61.02
2(Corn_N)	60.34	61.16	74.59	66.13	73.82	79.02
3(Corn_M)	59.85	59.67	82.40	70.89	68.23	84.80
4(Corn)	56.55	54.85	70.37	72.52	68.62	76.87
5(Grass_P)	35.26	57.41	75.93	80.91	78.07	90.89
6(Grass_T)	34.50	50.96	88.91	71.10	77.69	96.06
7(Grass_M)	26.13	18.03	48.89	33.85	72.00	52.78
8(Hay_W)	71.88	92.62	94.80	83.54	95.00	100
9(Oats)	21.43	51.85	34.15	38.89	33.33	50.00
10(Soybean_N)	56.40	59.47	83.84	79.02	81.58	89.15
11(Soybean_M)	76.95	71.61	84.19	79.82	86.16	86.51
12(Soybean_C)	49.53	61.99	75.22	68.05	61.68	65.75
13(Wheats)	65.13	74.53	74.53	75.97	93.69	88.08
14(Woods)	26.76	75.26	91.73	61.01	87.02	98.38
15(Building)	16.37	24.30	74.38	62.92	68.01	89.36
16(Tower)	42.23	50.60	52.10	85.57	75.24	67.74
0(Background)		89.51	92.18	87.47	86.59	86.76
$P_{OPR}(BKG)(\%)$	43.12	70.38	84.42	77.96	82.84	86.22
$P_{APR}(BKG)(\%)$	45.87	58.35	73.23	66.94	74.83	80.19
Time(s)	2448.4	287.3	291.6	286.5	288.4	293.7

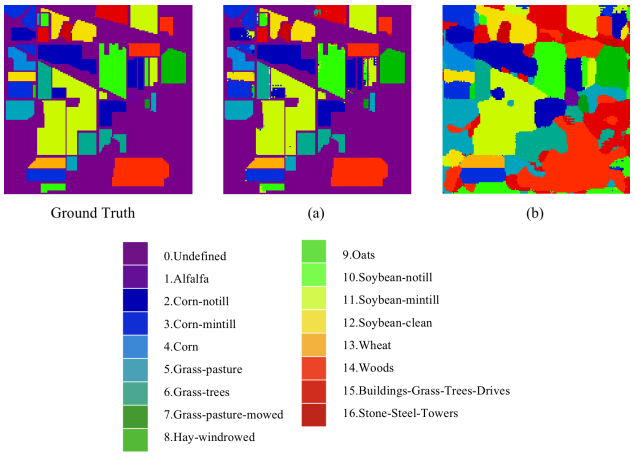


Fig. 16. Classification maps of SaaFormer for the Indian Pines dataset using 1% of training samples. (a) Without BKG. (b) With BKG.

TABLE XIII

P_A, P_{PR}, P_{OA}, AND P_{AA} OF IRTS-3D-CNN AND SAAFORMER USING 0.1% OF TRAINING SAMPLES AND $N_{bk}=0$ FOR THE SALINAS DATASET WITH THE HIGHEST ACCURACY BOLDFACED

	IRTS-3D-CNN		SaaFormer	
Class	$P_A(C_m)$	$P_{PR}(C_m)$	$P_A(C_m)$	$P_{PR}(C_m)$
1(Weeds_1)	99.35	68.62	99.95	**71.57**
2(Weeds_2)	**99.91**	63.77	99.73	**77.25**
3(Fallow)	96.58	**14.45**	**98.32**	11.08
4(Fallow_P)	**99.89**	**26.02**	98.92	15.02
5(Fallow_S)	98.82	**57.48**	**100**	49.10
6(Stubble)	99.42	**82.43**	**100**	73.11
7(Celery)	**99.94**	78.33	99.18	**82.17**
8(Grapes)	69.16	75.18	**80.40**	**91.02**
9(Soil)	99.60	28.60	**99.79**	**36.86**
10(Corn)	84.49	**64.65**	**93.80**	56.22
11(Lettuce_4)	**98.55**	19.41	98.22	**24.05**
12(Lettuce_5)	99.98	**32.37**	94.49	27.63
13(Lettuce_6)	98.47	60.09	**100**	**62.02**
14(Lettuce_7)	96.45	60.49	**98.78**	**76.17**
15(Vinyard_U)	**81.05**	57.44	73.71	**83.34**
16(Vinyard_T)	98.24	**55.08**	**99.11**	31.24
$P_{OA}(\%)$	89.60		**92.83**	
$P_{AA}(\%)$	94.99		**96.50**	
$P_{OPR}(BKG)(\%)$	43.65		**45.60**	
$P_{APR}(BKG)(\%)$	52.77		**53.40**	
Time(s)	5531.7		882.8	

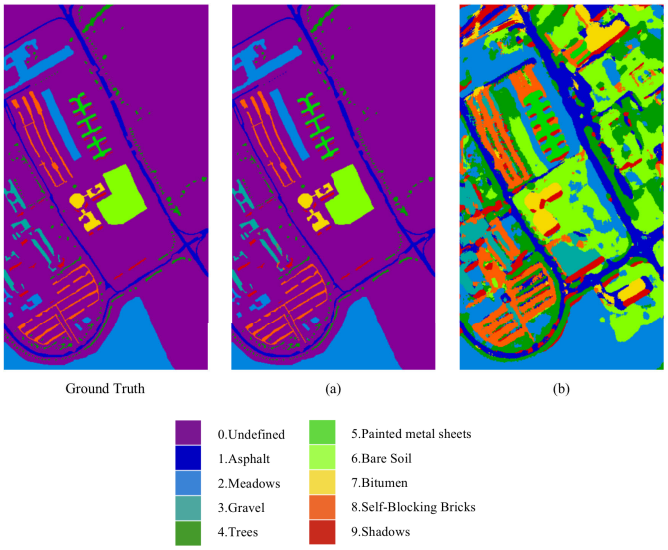


Fig. 17. Classification maps of SaaFormer for the PaviaU dataset using 0.1% of training samples. (a) Without BKG. (b) With BKG.

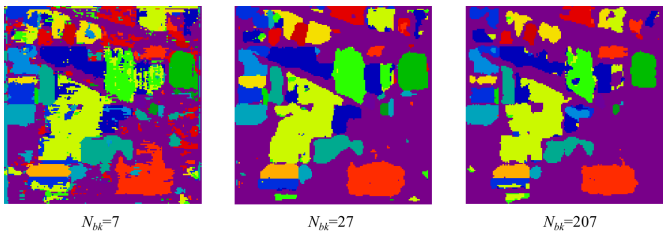


Fig. 18. Classification maps of SaaFormer for the Indian Pines dataset using different numbers N_{bk} of background samples and 1% training samples.

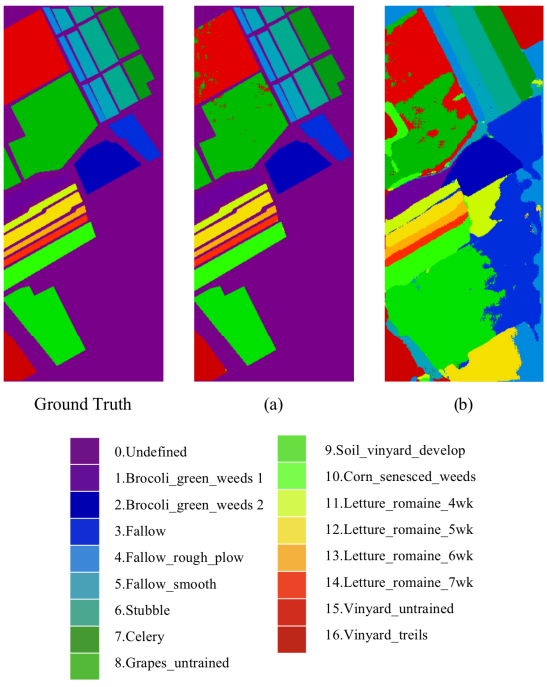


Fig. 19. Classification maps of SaaFormer for the Salinas dataset using 0.1% of training samples. (a) Without BKG. (b) With BKG.

is further segmented into $K=2$, $c_1=128$, and $c_2=32$ spectral clips, which is then processed through two consecutive transformer encoder blocks for classification. Each encoder block includes a four-head axial aggregation attention layer,

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TABLE XIV

P_A, P_{PR}, P_{OA}, AND P_{AA} OF IRTS-3D-CNN AND SAAFORMER USING 0.1% OF TRAINING SAMPLES AND $N_{\text{bk}} = 0$ FOR THE PAVIAU DATASET WITH THE HIGHEST ACCURACY BOLDFACED

	IRTS-3D-CNN		SaaFormer	
Class	$P_A(C_m)$	$P_{PR}(C_m)$	$P_A(C_m)$	$P_{PR}(C_m)$
1(Asphalt)	**86.77**	**23.44**	77.38	23.37
2(Meadows)	84.70	32.75	**88.42**	**43.19**
3(Gravel)	**93.67**	15.57	92.31	**16.27**
4(Tree)	**96.50**	**8.87**	96.34	6.56
5(Painted metal)	99.03	34.16	**100**	**44.22**
6(Bare Soil)	71.17	10.04	**91.60**	**16.34**
7(Bitumen)	**97.43**	**24.02**	97.36	17.73
8(Self-Blocking Bricks)	**95.38**	**15.41**	86.65	8.25
9(Shadows)	99.84	8.17	**100**	**12.38**
$P_{OA}(\%)$	86.82		**88.58**	
$P_{AA}(\%)$	91.61		**92.23**	
$P_{OPR}(BKG)(\%)$	17.91		**18.99**	
$P_{APR}(BKG)(\%)$	19.16		**20.92**	
Time(s)	4234.9		691.4	

TABLE XV

NUMBER OF TRAINING SAMPLES AND TEST SAMPLES ALLOCATED TO EACH CLASS FOR FOUR BENCHMARK DATASETS

Class	Indian Pines		Salinas		PaviaU		Houston	
	n_{train}	n_{test}	n_{train}	n_{test}	n_{train}	n_{test}	$n_{training}$	n_{test}
1	23	23	31	1978	56	6575	50	1201
2	36	1392	31	3695	56	18593	50	1204
3	35	795	31	1945	55	2044	50	647
4	32	205	31	1363	56	3008	50	1194
5	32	451	31	2647	56	1289	50	1192
6	35	695	31	3928	55	4974	50	275
7	14	14	31	3548	56	1274	50	1218
8	32	446	33	11238	55	3627	50	1194
9	10	10	32	6171	55	892	50	1202
10	36	936	32	3246			50	1177
11	43	2412	31	1037			50	1185
12	34	559	31	1896			50	1183
13	33	172	31	885			50	419
14	40	1225	31	1039			50	378
15	34	352	31	7237			50	610
16	31	62	31	1776				
BKG	-	10776	-	56975	-	164624	-	649816
Total	500	9749	500	53629	500	42276	750	14279

an MLP layer, and a GELU nonlinear activation layer. The specific parameters of our model have been detailed in the experimental setting, which employs an embedded spectrum with $d = 128$ channels. The time presented in Table XII indicates that our model exhibits relatively short execution times, with the size of N_{bk} exerting minimal influence on the runtime.

Furthermore, we compared and analyzed the experimental results with reference [69], which delved deeply into the background issue. For fairness, we adopt the same size of training samples for each class as [69], and the numbers of training samples and test samples for all image scenes are summarized in Table XV.

Figs. 20–23 show the classification maps produced by SaaFormer on the four datasets [69]. Meanwhile, Tables XVI–XIX summarize their averaged P_A, P_{PR}, P_{OA}, P_{AA}, P_{APR}, and P_{OPR}(BKG). At the same sampling rate, our model outperforms IRTS-NKCEM and IRTS-NKLCMV on Indian Pines, PaviaU, and Houston datasets for P_{OA}, P_{AA}, and P_{APR}. On the Salinas dataset, SaaFormer has a commendable performance with IRTS-NKCEM but is slightly worse than IRTS-NKLCMV.

TABLE XVI

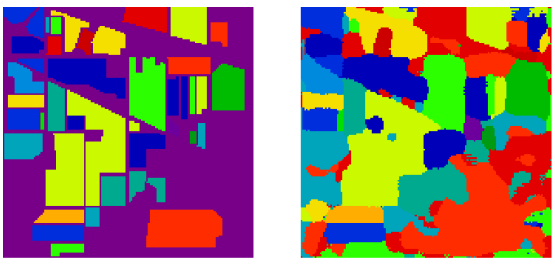
AVERAGED P_A, P_{PR}, P_{OA}, P_{AA}, P_{APR}, AND P_{OPR}(BKG) RESULTING FROM IRTS-NKCEM, IRTS-NKLCMV, AND SAAFORMER FOR THE INDIAN PINES DATASET

Class	IRTS-NKCEM		IRTS-NKLCMV		SaaFormer	
	$P_A(C_i)$	$P_{PR}(C_i)$	$P_A(C_i)$	$P_{PR}(C_i)$	$P_A(C_i)$	$P_{PR}(C_i)$
1	**100**	**100**	**100**	**100**	**100**	**100**
2	**98.96**	99.48	98.40	99.37	98.28	**99.71**
3	**99.81**	99.24	99.71	**99.43**	99.75	99.00
4	99.92	99.92	**100**	99.58	**100**	**100**
5	**98.72**	99.84	99.42	99.10	99.56	**100**
6	99.94	99.43	99.97	**99.81**	**100**	99.71
7	**100**	97.93	**100**	97.93	**100**	**100**
8	**100**	**100**	**100**	**100**	**100**	**100**
9	**100**	**100**	**100**	99.05	**100**	**100**
10	99.49	98.72	99.40	99.15	**99.57**	**99.36**
11	99.12	**99.67**	99.49	99.47	**99.79**	99.50
12	99.39	99.53	99.33	99.86	**99.64**	**100**
13	**100**	99.71	99.90	**100**	**100**	**100**
14	**100**	99.56	**100**	99.86	99.27	**99.84**
15	**100**	99.79	**100**	99.95	**100**	96.70
16	**100**	96.71	99.78	96.72	**100**	96.88
P_{OA}	99.48		99.50		**99.51**	
P_{AA}	99.71		99.71		**99.74**	
P_{APR}	99.34		99.32		**99.42**	
$P_{OPR}(BKG)$	48.49		**48.50**		47.31	

TABLE XVII

AVERAGED P_A, P_{PR}, P_{OA}, P_{AA}, P_{APR}, AND P_{OPR}(BKG) RESULTING FROM IRTS-NKCEM, IRTS-NKLCMV, AND SAAFORMER FOR THE SALINAS DATASET

Class	IRTS-NKCEM		IRTS-NKLCMV		SaaFormer	
	$P_A(C_i)$	$P_{PR}(C_i)$	$P_A(C_i)$	$P_{PR}(C_i)$	$P_A(C_i)$	$P_{PR}(C_i)$
1	**100**	**100**	**100**	**100**	**100**	**100**
2	**100**	**100**	**100**	**100**	**100**	**100**
3	**100**	**100**	**100**	**100**	**100**	**100**
4	99.80	**99.39**	99.94	98.94	**100**	99.27
5	**99.75**	99.90	99.49	99.97	99.62	**100**
6	99.99	**100**	99.98	99.99	**100**	**100**
7	**100**	99.99	**100**	99.98	**100**	**100**
8	97.89	98.17	**98.37**	**98.71**	94.36	97.92
9	**100**	99.96	**100**	**99.98**	**100**	99.87
10	99.91	**99.79**	**99.92**	99.75	99.75	99.54
11	**100**	**100**	**100**	**100**	**100**	**100**
12	**100**	**100**	99.99	**100**	99.95	**100**
13	99.98	**100**	99.96	99.98	**100**	99.89
14	99.87	**99.96**	99.85	99.81	**99.90**	99.90
15	97.13	96.81	**97.98**	**97.60**	96.89	91.89
16	**99.99**	**100**	99.98	**100**	99.94	**100**
P_{OA}	99.15		**99.35**		99.14	
P_{AA}	99.65		**99.72**		99.68	
P_{APR}	99.62		**99.67**		99.59	
$P_{OPR}(BKG)$	48.31		**48.40**		47.97	

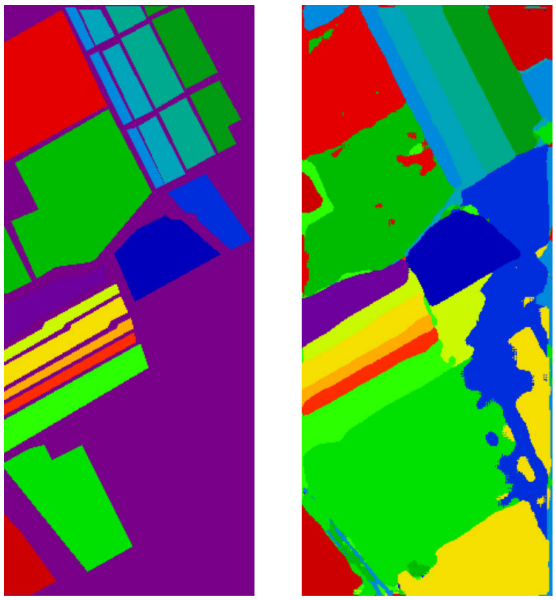


(a) (b)

Fig. 20. Final binary classification maps for the Indian Pines dataset produced by SaaFormer. (a) GT. (b) SaaFormer.

When computing P_{OPR}(BKG) values, it is essential to consider that the training set comprises labeled samples of known classes but excludes background samples. However, the test set includes background samples. Consequently, the model misclassifies all background samples as known labels,

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(a) (b)

Fig. 21. Final binary classification maps for the Salinas dataset produced by SaaFormer. (a) GT. (b) SaaFormer.

TABLE XVIII

AVERAGED P_A, P_{PR}, P_{OA}, P_{AA}, P_{APR}, AND P_{OPR}(BKG) RESULTING FROM IRTS-NKCEM, IRTS-NKLCMV, AND SAAFORMER FOR THE PAVIAU DATASET

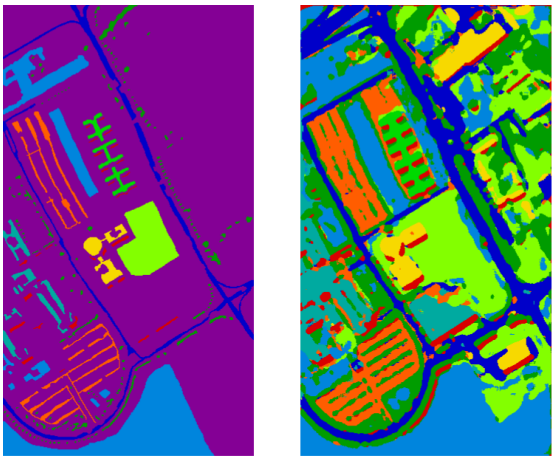
Class	IRTS-NKCEM		IRTS-NKLCMV		SaaFormer	
	$P_A(C_i)$	$P_{PR}(C_i)$	$P_A(C_i)$	$P_{PR}(C_i)$	$P_A(C_i)$	$P_{PR}(C_i)$
1	98.86	99.51	98.73	99.50	**98.87**	**99.63**
2	99.37	99.67	99.52	99.77	**99.78**	**99.93**
3	94.75	**98.88**	95.93	98.43	**96.18**	96.75
4	98.53	98.49	**98.90**	98.22	98.53	**99.60**
5	**100**	99.99	**100**	99.99	**100**	**100**
6	99.68	98.52	99.73	**99.31**	**99.80**	99.08
7	**99.79**	**100**	**99.79**	**100**	**100**	97.70
8	**98.63**	95.31	98.42	95.57	98.59	**97.15**
9	99.89	**100**	99.91	99.96	**100**	99.89
P_{OA}	99.02		99.14		**99.29**	
P_{AA}	98.83		98.99		**99.40**	
P_{APR}	98.93		98.97		**99.07**	
$P_{OPR}(BKG)$	20.44		**20.45**		20.37	

TABLE XIX

AVERAGED P_A, P_{PR}, P_{OA}, P_{AA}, P_{APR}, AND P_{OPR}(BKG) RESULTING FROM IRTS-NKCEM, IRTS-NKLCMV, AND SAAFORMER FOR THE HOUSTON DATASET

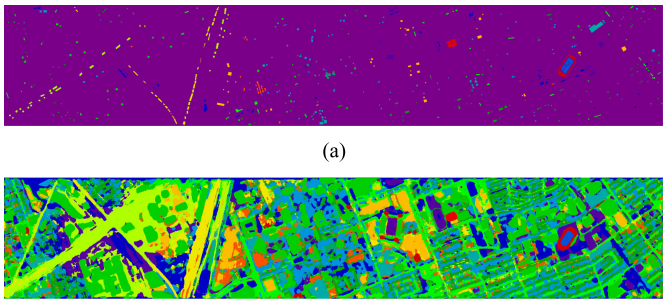
Class	IRTS-NKCEM		IRTS-NKLCMV		SaaFormer	
	$P_A(C_i)$	$P_{PR}(C_i)$	$P_A(C_i)$	$P_{PR}(C_i)$	$P_A(C_i)$	$P_{PR}(C_i)$
1	99.45	98.85	99.18	**99.24**	**99.83**	98.12
2	99.31	**99.88**	**99.49**	99.81	98.33	99.66
3	99.37	**100**	**99.92**	99.97	99.84	**100**
4	**99.92**	99.46	99.90	99.24	99.66	**99.92**
5	99.95	99.92	99.44	**100**	**100**	**100**
6	93.51	**100**	92.10	**100**	**100**	**100**
7	98.78	97.22	**99.10**	97.50	**99.10**	**98.29**
8	**98.45**	98.27	97.46	99.31	96.73	**99.91**
9	**97.79**	98.09	97.73	96.19	97.08	**98.81**
10	**99.91**	**98.85**	99.65	98.12	98.89	95.88
11	98.21	**98.72**	97.61	97.61	**98.31**	97.90
12	**98.82**	98.03	97.27	98.06	98.73	**99.57**
13	**96.39**	**96.98**	96.04	96.89	95.47	92.81
14	98.87	**100**	**100**	**100**	**100**	**100**
15	99.65	99.62	99.97	**99.92**	**100**	99.35
P_{OA}	98.79		98.59		**99.04**	
P_{AA}	98.50		98.33		**99.19**	
P_{APR}	98.92		98.79		**99.09**	
$P_{OPR}(BKG)$	**2.57**		2.56		2.13	

leading to the $P_{(\text{OPR})}$(BKG) value decreasing when the size of the background samples increases. Theoretically, by (17) and (18), P_{OPR}(BKG) is proportional to P_{APR}. It is feasible



(a) (b)

Fig. 22. Final binary classification maps for the PaviaU dataset produced by SaaFormer. (a) GT. (b) SaaFormer.



(a)

(b)

Fig. 23. Final binary classification maps for the Houston dataset produced by SaaFormer. (a) GT. (b) SaaFormer.

TABLE XX

$P_{\text{OPR}}(\text{BKG})_{\max}$ AND P_{OPR}(BKG) FOR THE FOUR DATASETS PRODUCED BY SAAFORMER

	IndianPines	Salinas	PaviaU	Houston
$P_{PR}(BKG)_{max}(\%)$	47.50	48.49	20.43	2.15
$P_{PR}(BKG)(\%)$	47.31	47.97	20.37	2.13
$\frac{P_{PR}(BKG)}{P_{PR}(BKG)_{max}}(\%)$	99.60	98.93	99.61	99.07

to calculate the theoretical upper bound of the P_{OPR}(BKG) for these four datasets, i.e., $P_{\text{OPR}}(\text{BKG})_{\max} = (n_{\text{test}}/n_{\text{test}} + n_{\text{BKG}})$, where n_{BKG} represents the total number of background samples. Table XX displays the theoretical upper bounds of the P_{OPR}(BKG) values for these datasets alongside the results obtained by the model. Examination of Table XX reveals that although the P_{OPR}(BKG) values are relatively low, they are remarkably close to their theoretical upper bounds.

V. CONCLUSION

HSIs are often represented as data cubes with both spatial and spectral dimensions, typically viewed sequentially along the spectral axis. Deep neural networks have achieved high accuracy on these images even with limited training data, primarily under the commonly used random sampling evaluation method. However, we found a significant overlap between the training and validation datasets, with classification

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accuracy directly related to this overlap. This suggests that random sampling overestimates the model's generalization ability. To address this, we propose nonoverlapping data partitioning methods for more accurate generalization assessment. Our experiments show that popular models, including 2DCNN, 3DCNN, RNN, and transformer-based models, experience notable performance drops with nonoverlapping partitions. In response, we introduce SaaFormer, a transformer-based model that emphasizes spectral characteristics through axial aggregation attention and multilevel spectral extraction. SaaFormer maintains similar performance under random sampling but significantly outperforms other methods with alternative sampling strategies, as demonstrated on five public datasets.

Our future work will focus on developing more generalizable and interpretable models that integrate deep learning with traditional methodologies. The primary emphasis will be on enhancing model interpretability and accurately and efficiently determining the optimal number of layers in multilevel spectral extraction. To achieve this, we will incorporate the physical characteristics of spectral bands and prior knowledge of hyperspectral images to construct models with stronger interpretative capabilities. In addition, we will explore the embedding of interpretative knowledge as a means to develop and evaluate more advanced models.

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Enzhe Zhao (Student Member, IEEE) received the master's degree from Harbin Institute of Technology, Harbin, China, in 2022, where she is currently pursuing the Ph.D. degree with the School of Mathematics.

Her research interests include image processing and analysis, hyperspectral remote sensing, and deep learning.

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Zhichang Guo received the Ph.D. degree from Jilin University, Changchun, China, in 2010.

He is currently a Professor with the School of Mathematics, Harbin Institute of Technology, Harbin, China. His research interests include partial differential equations, nonlinear analysis, mathematical methods in image analysis, and deep learning.



Shengzhu Shi received the Ph.D. degree from Harbin Institute of Technology, Harbin, China, in 2022.

She is currently a Lecturer with the School of Mathematics, Harbin Institute of Technology. Her research interests include numerical methods for partial differential equations, image processing, deep learning, and uncertainty quantification.



Yao Li (Member, IEEE) received the Ph.D. degree in system engineering from the University of Illinois Urbana–Champaign, Champaign, IL, USA, in 2020.

He is currently an Assistant Professor with the School of Mathematics, Harbin Institute of Technology, Harbin, China. His research interests include adversarial attacks, image processing, and electroencephalogram (EEG) processing.



Jia Li received the Ph.D. degree from Harbin Institute of Technology, Harbin, China, in 2020.

He is currently working as an Assistant Professor at the School of Mathematics, Harbin Institute of Technology. His research interests include numerical methods for partial differential equations and deep learning.



Dazhi Zhang received the Ph.D. degree from Jilin University, Changchun, China, in 2006.

He is currently a Professor with the School of Mathematics, Harbin Institute of Technology, Harbin, China. His research interests include partial numerical solutions of partial differential equations, mathematical methods in image analysis, deep learning, and remote sensing.

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